INCH-POUND

MIL-DTL-52286D
18 May 2010
SUPERSEDING
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28 April 1980

DETAIL SPECIFICATION

CABLE ASSEMBLIES, POWER, ELECTRICAL (WITH MOLDED ON TERMINATIONS), GENERAL SPECIFICATION FOR

Inactive for new design after 9 December 2003.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cable assemblies with one or more molded-rubber terminals for interior and exterior power and lighting.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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FEDERAL SPECIFICATION

TT-C-490	- Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

COMMERCIAL ITEM DESCRIPTION

A-A-2917	- Twine, Cotton, Seine

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-3432	- Cables, (Power and Special Purpose) and Wire, Electrical (300 and 600 Volts)
MIL-I-3930	- Insulating and Jacketing Compounds, Electrical (for Cables, Cords, and Wires), General Specification for
DOD-P-15328	- Primer (Wash), Pretreatment (Formula No. 117 for Metals) (Metric)
MIL-DTL-15659	- Terminal, Lug: Solder, Copper and Phosphor Bronze
MIL-I-24391	- Insulation Tape, Electrical, Plastic Pressure-Sensitive

(See supplement 1 for list of specification sheets.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-130	- Identification Marking of U.S. Military Property

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-454	- General Guidelines for Electronic Equipment

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from http://www.asq.org/ or American Society of Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

ASTM INTERNATIONAL

ASTM A466/A466M	- Standard Specification for Weldless Chain
ASTM B26/B26M	- Standard Specification for Aluminum-Alloy Sand Castings
ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D2240	- Standard Test Method for Rubber Property - Durometer Hardness

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

MASTER PAINTERS INSTITUTE (MPI)

Reference Number 8	- Alkyd, Exterior Flat (MPI Gloss Level 1)
Reference Number 9	- Alkyd, Exterior Gloss (MPI Gloss Level 6)
Reference Number 47	- Alkyd, Interior, Semi-Gloss (MPI Gloss Level 5)
Reference Number 48	- Alkyd, Interior, Gloss (MPI Gloss Level 6)
Reference Number 49	- Alkyd, Interior, Flat (MPI Gloss Level 1)
Reference Number 51	- Alkyd, Interior, (MPI Gloss Level 3)
Reference Number 94	- Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)

(Copies of these documents are available from http://www.paintinfo.com/mpi/ or from Master Painters Institute, 2800 Ingleton Ave., Burnaby, BC V5C 6G7, Canada.

SAE INTERNATIONAL

SAE AS7928	- Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for

(Copies of this document are available from http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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SOCIETY FOR PROTECTIVE COATINGS

SSPC-Paint 25 - Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel Type I and Type II

(Copies of this document are available from http://www.sspc.org/ or from The Society for Protective Coatings, 40 24th Street, 6th Floor, Pittsburgh, PA 15222-4656.)

UNDERWRITERS LABORATORIES

UL 496 - Lampholders

(Copies of these documents are available from http://www.ul.com/ or from Underwriter Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Description. Cable assemblies shall consist of cables, connectors, dummy connectors, adapters, and terminal lugs and shall be as shown on figures 1 through 46 and applicable specification sheets. Dimensions and tolerances are shown where required. Where tolerances are not shown or where those prescribed could result in incorrect fits, the contractor is responsible for providing tolerances to insure correct fit, assembly, and operation of each assembly. No deviation from dimensions or tolerances shown on the figures is permissible without prior approval of the contracting officer.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3.1 First article sample. The contractor shall furnish two cable assemblies of each type supplied for examination and test within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce cable assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests (see 6.2).

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3.4 Operation. Each cable assembly, when connected to a source of power and carrying rated current shall not overheat or show signs of arcing. The connectors or terminals shall show no signs of electrical or mechanical malfunction or damage when connected or disconnected.

3.5 Cable. The cable shall be of the low-temperature, heat-resistant kind, conforming to MIL-DTL-3432 and shall be of the type and length specified on the applicable specification sheet.

3.6 Molded connectors (plugs, receptacles, inserts, adapters, and dummy connectors). Material used for insulation and jacketing shall be the same as the cable to which the connectors are to be vulcanized and shall have a durometer type A hardness of 60 ±10 as described in ASTM D2240. The insulation shall fully enclose the base of the pins and sockets. The jacket shall be molded integrally to the insulation. The connector shall be vulcanized to the cable and shall form a watertight seal as defined in MIL-STD-108.

3.7 Metal connector with molded insert.

3.7.1 Shell and cover. Connector shells and covers for molded inserts shall be of cast aluminum conforming to ASTM B26/B26M, alloy 356, temper T6. Pins and nuts used to secure the covers shall be of carbon steel.

3.7.2 Removability of inserts. All molded inserts for connectors shall be removable. Provisions shall be made to position inserts to prevent rotation with respect to the shell.

3.8 Dummy connectors. Dummy connectors shall be molded from insulation material conforming to MIL-I-3930 and as specified on the applicable figure. Dummy connectors shall prevent the entrance of moisture, dirt, or other foreign material into the specified plug.

3.8.1 Attaching means. Dummies shall be attached to the cable by either a length of cotton twine or a brass chain as specified on the applicable specification sheet.

3.8.1.1 Cotton twine. Cotton twine shall conform to A-A-2917, Size No. 18, and shall be attached to the dummy and cable assembly by means of a durable knot which will not slip when pulled.

3.8.1.2 Chain. Chain shall conform to ASTM A466/A466M, plumber's chain (class SF), trade size 1/0; nominal metal thickness 0.023 inch; approximately 24 links per foot; and minimum breaking force of 110 pounds.

3.9 Connector adapters. Connector adapters shall consist of plugs and receptacles molded together to form a single piece. The material used for insulation shall be IS-L and jacketing shall be JS-L in accordance with MIL-I-3930. The insulation shall fully enclose the base of the pins and sockets. The jacket shall be molded integrally to the insulation. Three-way adapters shall be furnished with two dummy connectors; five-way adapters shall be furnished with four dummy connectors.

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3.10 Caps. A watertight cap shall be placed on each single-conductor plug and each single-conductor receptacle for protection during shipment and installation. The cap shall consist of a rubber plug or receptacle made of jacket material conforming to MIL-I-3930 and shall be designed to protect the mating surfaces of the connector from moisture and dirt. Tape shall be wrapped over the joining seam to hold the cap securely in place and to give additional protection from moisture and dirt. The tape shall conform to MIL-I-24391 and shall be 0.0085 ±0.001 inch thick and 0.75 inch wide. The cap shall not damage or deform the connector in any way.

3.11 Pins and sockets. Pins and sockets of the molded connectors, inserts, and adapters shall be centered, anchored, and molded perpendicular to the face of the component so that they will not loosen by repeated connection and disconnection with other electrical connectors. There shall be no intrusion of insulation or foreign material into the socket. Contact dimensions shall be as specified in table I. The illustrations shown on figure 1 are for dimensional purposes only and are not intended to indicate the design. Applicable type numbers of contacts shall be as specified on the applicable figure.

3.11.1 Pins. The entering end of the pin shall be hemispherical with a radius of approximately one-half of the pin diameter.

3.11.2 Sockets. The receiving end of the socket shall be rounded or chamfered for directing and centering the entering pin. Sockets shall have three slots 0.02 inch maximum width and approximately 120 degrees apart and shall have circumferential-type retaining springs for maintaining positive electrical contact between pin and socket. Sockets shall have an external metallic retaining sleeve or band to prevent distortion of the socket.

3.11.3 Wire connections. Each conductor shall be mechanically and electrically bonded to the pin or socket. Each shall be closed at the end connected to the conductor to prevent moisture from entering the cable.

3.11.3.1 Strain relief. The conductors within the connector shall be arranged with adequate slack to prevent the conductors from breaking, or separating from pins or sockets, when subjected to jerks or pulls on the cable.

3.12 Bond between connectors and cables. The bond between a connector or insert and a cable shall withstand the pull specified on the applicable specification sheet for 1 minute without mechanical or electrical damage.

3.13 Degree of closure. Molded connectors, adapters, inserts, caps, and dummy connectors when mated with applicable mating components shall provide a watertight joint as defined in MIL-STD-108.

3.14 Connect and disconnect force. Connect and disconnect force of the connectors shall be as shown on the applicable figure and connectors shall be constructed to withstand 500 repeated connections and disconnections without damage.

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TABLE I. Contact dimensions.

TYPE		Pin A (inch)	Pin B (inch)	Socket C (inch)	Socket D (inch)
I	max.	0.656	0.156	0.703	0.158
	min.	0.594	0.154	0.641	0.156
II S	max.	1.109	0.187	1.172	0.189
	min.	1.031	0.185	1.094	0.187
II P	max.	1.109	0.250	1.172	0.252
	min.	1.031	0.248	1.094	0.250
III S[1]	max.	1.250	0.312	1.312	0.314
	min.	1.125	0.310	1.188	0.312
III P[2]	max.	1.250	0.375	1.312	0.377
	min.	1.125	0.373	1.188	0.375
IV S	max.	1.641	0.375	1.750	0.377
	min.	1.562	0.486	1.625	0.375
IV P	max.	1.641	0.438	1.750	0.440
	min.	1.562	0.436	1.625	0.438
V	max.	1.077	0.187	1.125	0.189
	min.	1.015	0.185	1.062	0.187
VI S	max.	0.828	0.156	0.891	0.158
	min.	0.750	0.154	0.859	0.156
VI P	max.	0.828	0.218	0.891	0.220
	min.	0.750	0.216	0.859	0.218
VII S	max.	0.515	0.125	0.562	0.127
	min.	0.453	0.123	0.500	0.125
VII P	max.	0.515	0.156	0.562	0.158
	min.	0.453	0.154	0.500	0.156
VIII S	max.	0.490	0.125	-	-
	min.	0.360	0.123	-	-
VIII P	max.	0.490	0.156	-	-
	min.	0.360	0.154	-	-
IX	max.	1.703	0.312	1.781	0.314
	min.	1.601	0.310	1.703	0.312
X S	max.	0.640	0.125	0.703	0.127
	min.	0.610	0.123	0.641	0.125
X P	max.	0.640	0.156	0.703	0.158
	min.	0.610	0.154	0.641	0.156
XI	max.	2.250	0.625	2.312	0.627
	min.	2.187	0.623	2.250	0.625

[1] S denotes standard contact
[2] P denotes polarizing contact

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3.15 Voltage drop. The voltage drop across the terminals of mating contacts shall not exceed the maximum specified on the applicable figure.

3.16 Dielectric strength. Molded components shall withstand 5 times their rated voltage for 1 minute without damage, arcing or breakdown.

3.17 Cable hangers. Cable hangers for cable assemblies and extension lights shall be molded of the same jacket material as the cable and shall be securely vulcanized to the cable jacket.

3.18 Lampholders. Lampholders shall conform to UL 496 and be of the cap and shell type, with either a brass or steel shell and matching pendant cord cap with integrated strain relief. The shell shall include threads to accommodate a shadeholder. A continuous circular bead or shoulder shall be provided directly above the shadeholder thread for use as a stop for the shadeholder. The pendant cap shall include a composition bushing and cord grip sized to accommodate a cord diameter of 0.250 in. to 0.375 in. The lampholder shall be rated for 250 volts alternating current (VAC), 250 watts.

3.19 Terminal lugs. Solder type terminal lugs shall conform to MIL-DTL-15659; solderless terminal lugs shall conform to SAE AS7928. Solder type terminal lugs, when used, shall be soldered to the wire in accordance with MIL-HDBK-454.

3.20 Bolts, washers and nuts. The bolts, washers, and nuts shall be carbon steel, zinc plated in accordance with ASTM B633, service condition 2 (SC 2), type II.

3.21 Temperature and humidity. The cable assembly shall not be damaged by operation, carrying rated voltage and current, or storage at ambient temperatures of +125 °F and -65 °F, and at maximum humidity possible at these temperatures.

3.22 Treatment and painting. Finish of metal parts not otherwise specified on the figures shall be cleaned, pretreated, primed and painted as described in the following paragraphs. Surface blasting of steel surfaces shall not be required. Carbon steel shall be zinc plated in accordance with ASTM B633, SC 2, type I, prior to treatment and painting.

3.22.1 Cleaning. Unless otherwise specified in the end item specification or elsewhere (see 3.22.5), cleaning procedures shall be in accordance with TT-C-490 or via government-approved industry best practices, including:

a. Chemical methods such as solvent cleaning, alkaline cleaning, acid cleaning, pickling, descaling with hydride, or paint stripping.

b. Electrochemical cleaning methods, such as electrolyte alkaline, electropolishing, or electrolytic pickling.

After cleaning, all surfaces shall be kept free from dirt, dust, finger marks, and other contaminants, until treated as specified in 3.22.2.

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3.22.2 Pretreatment. Once cleaned, parts shall be pretreated in accordance with DOD-P-15328 using a coating thickness of 0.3 to 0.5 mils.

3.22.3 Priming. Pretreated parts shall be primed in accordance with SSPC-Paint 25 using a coating thickness of 1.0 to 1.5 mils.

3.22.4 Topcoating. Primed parts shall be topcoated to a thickness of 1.0 to 1.5 mils using the appropriate topcoat as specified in table II. Users will evaluate replacements for their particular application.

TABLE II. MPI reference numbers.

No.	Name	Gloss level
8	Alkyd, Exterior Flat	1
9	Alkyd, Exterior Gloss	6
47	Alkyd, Interior, Semi-Gloss	5
48	Alkyd, Interior, Gloss	6
49	Alkyd, Interior, Flat	1
51	Alkyd, Interior,	3
94	Alkyd, Exterior, Semi-Gloss	5

3.22.5 Hardware and hardware items. Hardware and hardware items such as bolts, capscrews, washers, pins, springs and other fittings are not to be cleaned and pretreated prior to assembly and painting if there is no evidence of rust or corrosion.

3.23 Identification. In addition to any marking specified in the applicable figure, the cable assemblies shall be permanently and legibly marked in accordance with MIL-STD-130 and shall include date of manufacture. On cable assemblies, the marking shall be applied at each end. Marking shall be in the form of a durable band which shall encircle and adhere to the cable. Marking shall be clear and fused into the backing. Both the band and its marking shall show no indication of cracking upon repeated flexing of the cable and shall be resistant to moisture and oil.

3.24 Workmanship. The cable assemblies shall be free from defects such as molding flaws, tear through connector skirt or cork, split seam at joint between cable and connector, connectors not properly mating, or other defects that would impair use of the assembly.

3.25 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. Each first article assembly shall be examined in accordance with table III. Presence of one or more defects in either first article cable assembly shall be cause for rejection of both the assemblies.

4.2.2 Tests. Following successful completion of the examination, the first article cable assemblies shall be tested as indicated in table IV. Tests shall be conducted in the order listed. Any test result not meeting the applicable requirement paragraph shall constitute failure of the test and shall be cause for rejection of the first article cable assemblies.

4.3 Conformance inspection.

4.3.1 Sampling. Sampling for tests shall be in accordance with ASQ Z1.4, with the Acceptance Quality Limit (AQL) as specified in the contract (see 6.2). Inspection level S-2 shall be used for lots of 2 to 65. For lots of 66 and larger, inspection level S-3 shall be used.

4.3.2 Examination.

4.3.2.1 Individual. Each production cable assembly shall be examined in accordance with table III. Presence of one or more defects shall be cause for rejection.

4.3.3 Test.

4.3.3.1 Individual. Each production cable assembly shall be tested as indicated in table IV. Failure of any test shall be cause for rejection.

4.3.3.2 Sample. Samples selected in accordance with 4.3.1 that have passed the individual test specified in 4.3.2.1 shall be tested in accordance with table IV.

4.4 Inspection procedure.

4.4.1 Inspection schedule. Tables III and IV shall be used for first article and conformance inspections as specified in 4.3 and 4.4.

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TABLE III. Examination.

Cable assembly	Adapter	Dummy connector	Characteristics	Requirement paragraph
X	X	-	Pins or socket contacts not as specified.	3.2
X	X	X	Dimensions not as specified.	3.2
X	-	-	180° bend at joint causing visible separation of connector and cable.	3.2
X	X	X	Materials or components not as specified.	3.2
X	-	-	Gap between skirts exceeds .063 inch but less than .125 inch when mated.	3.2, 3.13
X	-	-	Bond between cable and connectors not watertight.	3.6
X	X	X	Attaching means for dummy connectors not as specified.	3.8.1
X	X	-	Rubber in socket adheres to contact area or filling slit.	3.11
X	X	-	Connectors not electrically or physically serviceable after mating.	3.11.2
X	X	-	Lampholders not as specified.	3.18
X	-	-	Treatment and painting not as specified.	3.22
X	X	X	Identification marking missing or incomplete.	3.23
X	X	X	Workmanship not as specified.	3.24

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TABLE IV. Test schedule.

First article			Conformance						Test	Test paragraph	Req. paragraph
			Cable assembly		Adapter		Dummy connector				
Cable assembly	Adapter	Dummy connector	Ind.	Samp.	Ind.	Samp.	Ind.	Samp.			
X	X	-	X	-	X	-	-	-	Operating	4.5.1	3.4
X	-	-	-	-	-	-	-	-	Cable	4.5.2	3.5
X	X	X	-	-	-	-	-	-	Insulation and jacket	4.5.3.1	3.6
X	-	-	-	-	-	-	-	-	Inserts	4.5.3.2	3.6
X	-	-	-	X	-	-	-	-	Bond between connector and cable	4.5.4	3.12
X	X	X	-	X	-	X	-	X	Degree of closure	4.5.3.3.1	3.13
X	X	-	-	-	-	-	-	-	Connect and disconnect force	4.5.3.3.3	3.14
X	X	-	-	-	-	-	-	-	Voltage drop	4.5.5	3.15
X	X	-	-	X	-	X	-	-	Dielectric strength	4.5.3.3.2	3.16
X	X	X	-	-	-	-	-	-	Temperature and humidity	4.5.6	3.21

4.5 Test procedures.

4.5.1 Operating. Connect the cable assembly to a power source and dummy load, and check for operation at rated voltage and current. Connect and disconnect the connectors five times. Short circuits, open circuits, arcing during operation, or failure of the connector to connect or disconnect shall constitute failure of this test.

4.5.2 Cable. Test the cable in accordance with MIL-DTL-3432 prior to fabrication into an assembly, or the cable manufacturer shall provide a certification that the cable meets the requirements of MIL-DTL-3432.

4.5.3 Molded connector, adapter, inserts, dummies, and plugs.

4.5.3.1 Insulation and jacket. Test the insulation and jacket material used in fabrication of the connectors, adapters, inserts, dummies, and plugs in accordance with MIL-I-3930, or the manufacturer of the material shall provide a certification that the cable meets the requirements of MIL-DTL-3432.

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4.5.3.2 Hardness of insert or body material. Determine the hardness of the body material using procedures for type A durometer testing as described in ASTM D2240.

4.5.3.3 Degree of closure and dielectric strength.

4.5.3.3.1 Degree of closure. Install a mating plug on not less than one accommodation of the cable-assembly terminal. The mating connector plug shall be equipped with a minimum of three feet of cable with bared ends. Cap all additional cable-assembly terminals with mating plugs or connector dummies. When adapters are required, use two connector plugs for mating with the adapter. Cap all other accommodations of the adapter with dummy connectors. Keeping the bared ends dry, immerse each assembly to a depth of three feet in tap water for a period of eight hours. Any evidence of moisture in the interior of either the dummy connector or its mating connector at the end of eight hours shall constitute failure of this test.

4.5.3.3.2 Dielectric strength. Apply the test voltage to the adjacent conductors. Watertight assemblies shall be tested immersed in water with the voltage applied between each conductor and the grounded water bath in which the cable is immersed. Supply the test voltage from an alternating-current source with a frequency of not more than 100 cycles per second. Apply the nominal-use voltage initially, increase at an approximately uniform rate to five times the rated voltage, and hold at that value for one minute. The rate of voltage increase shall not be less than 100 percent in 60 seconds. Measure the test voltage by any method that gives a root-mean-square (rms) value. A crest factor of the test voltage, if different, by more than ±10 percent from that of a sinusoidal wave, when the source is loaded with the test specimen, shall constitute failure of this test.

4.5.3.3.3 Connect and disconnect force. Determine the force required to connect and disconnect the connector plug of the cable assembly with a mating connector as follows: Mount the test item in a manner so that a static load maybe applied along the longitudinal axis of the test item, and gradually increase the static load until the mating connector is engaged to the depth required for normal operation. Determine the force required to disconnect the mating connector from the test item by applying a gradually increasing longitudinal pull on the mating connector. Conduct the test 500 times. Examine and operate the connectors at rated voltage and current. Any damage that will impair use shall constitute failure of this test.

4.5.4 Bond between connectors and cables. The connector body shall be rigidly supported and a weight or force shall be gradually applied to the cable. Apply the maximum weight or force specified on the applicable figure for one minute. Check the cable assembly during the test for continuity of circuit. Any evidence of tearing, breakage or other damage that would impair use shall constitute failure of this test.

4.5.5 Voltage drop. Determine the voltage drop between mated pins and sockets by applying a direct current (DC) voltage at the current stipulated in the applicable figure and measuring between the ends of the terminals. Voltage drop exceeding the limit specified in 3.15 shall constitute failure of this test (see figure 1).

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4.5.6 Temperature and humidity. Maintain chambers used for temperature and humidity tests within ±4 °F of the specified temperature throughout the test.

4.5.6.1 Low-temperature storage and operation. Install a mating plug on each end of the cable assembly that is provided with a plug. Equip each mating plug with a minimum of three feet of cable with bared ends. Place the cable assemblies in a test chamber and expose to an ambient temperature of -65 °F, for 24 hours. Test each assembly as specified in 4.5.1. Any defect in connectors, adapters, dummies, or plugs such as cracks or separation of connectors from the cable shall constitute failure of this test.

4.5.6.2 High-temperature and humidity storage and operation. Install a mating plug on each end of each cable assembly provided with a plug. Equip the mating plug with a minimum of three feet of cable. Place the cable assemblies in the test chamber and expose to an ambient temperature of 125 °F, and maximum humidity for a period of 24 hours. Test each assembly as specified in 4.5.1. Any defects in connectors, adapters, dummies, or plugs such as cracks or separation of connectors from the cable shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cable assemblies are intended for use in electrical distribution systems requiring all-weather capability.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable specification sheet (see 3.1).
c. When first article is required (see 3.3).
d. Time frame required for submission of the first article samples (see 3.3.1).
e. AQL (see 4.3.1).
f. Packaging requirements (see 5.1).

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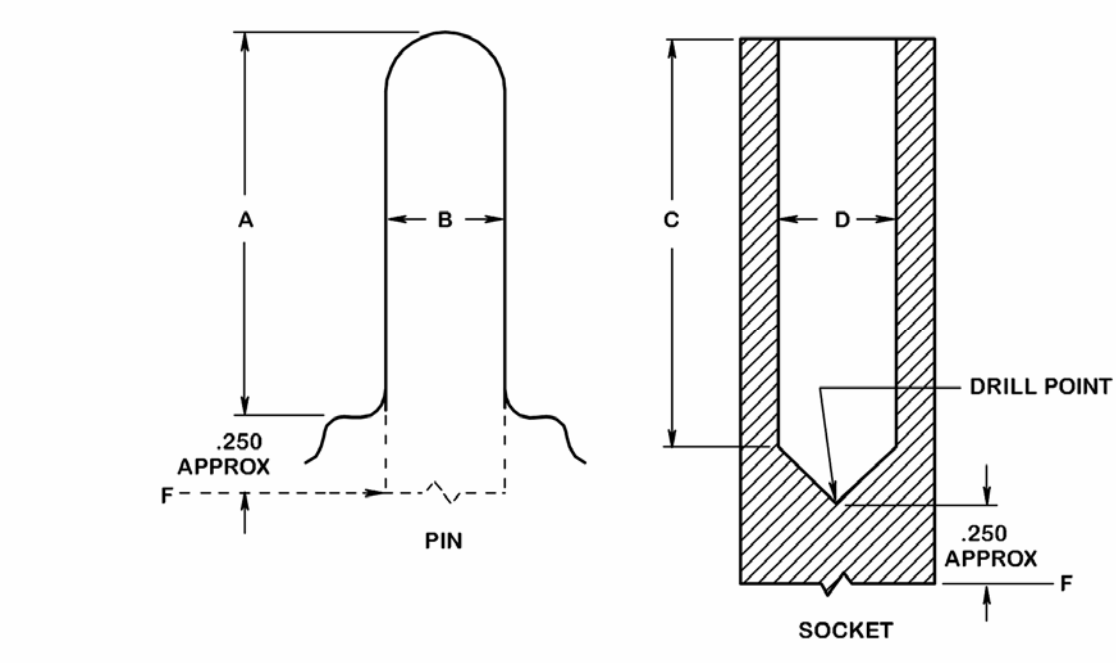
6.3 First article sample. Any changes or deviations of production cable assemblies from the approved first article samples during production will be subject to the approval of the contracting officer. Approval of the first article samples will not relieve the contractor of his obligation to furnish cable assemblies conforming to this specification.

6.4 Subject term (key word) listing.

Adapter
All weather
Connector
Cover
Distribution
Dummy
Exterior
Insert
Insulation
Jacketing
Lighting
Pin
Plug
Receptacle
Shell
Socket
Terminal
Weatherproof

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

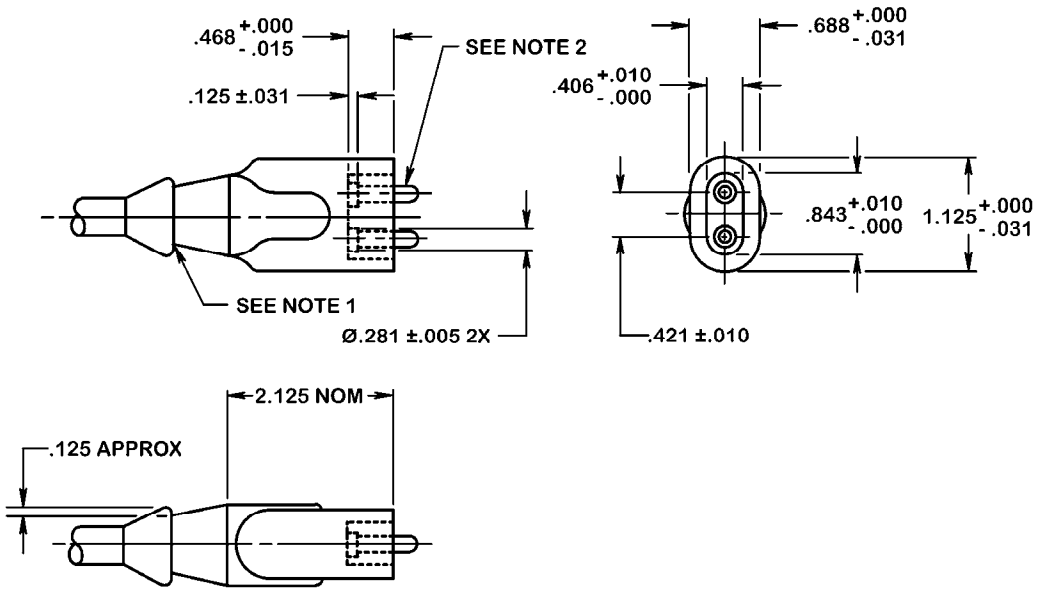
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NOTES:
1. F is the point for voltage drop probe on contact.
2. All dimensions are in inches.

FIGURE 1. Contacts.

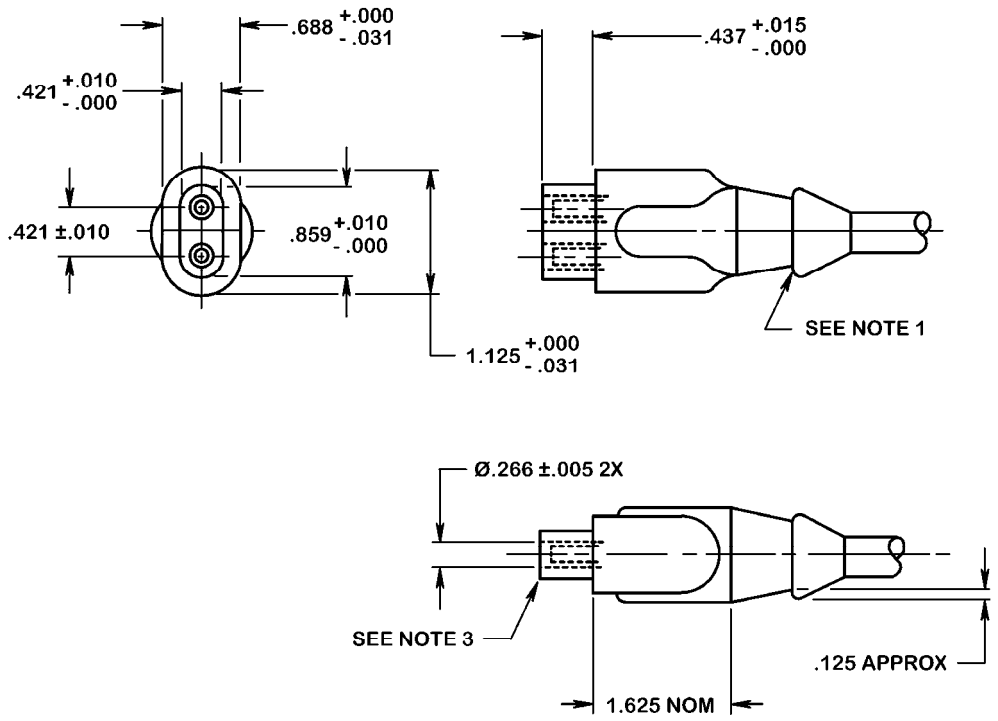
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NOTES:
1. This groove to be of sufficient depth, width, or cross section to be used with twine or cord of .125 diameter and need not be of the shape shown.
2. Type I male contacts.
3. Connection rating: 600 volts, 15 amperes.
4. Voltage drop: Not more than 10 millivolts at 15 amperes.
5. Connecting and disconnecting force: 10-35 pounds.
6. All dimensions are in inches.

FIGURE 2. Plug, molded, (male).

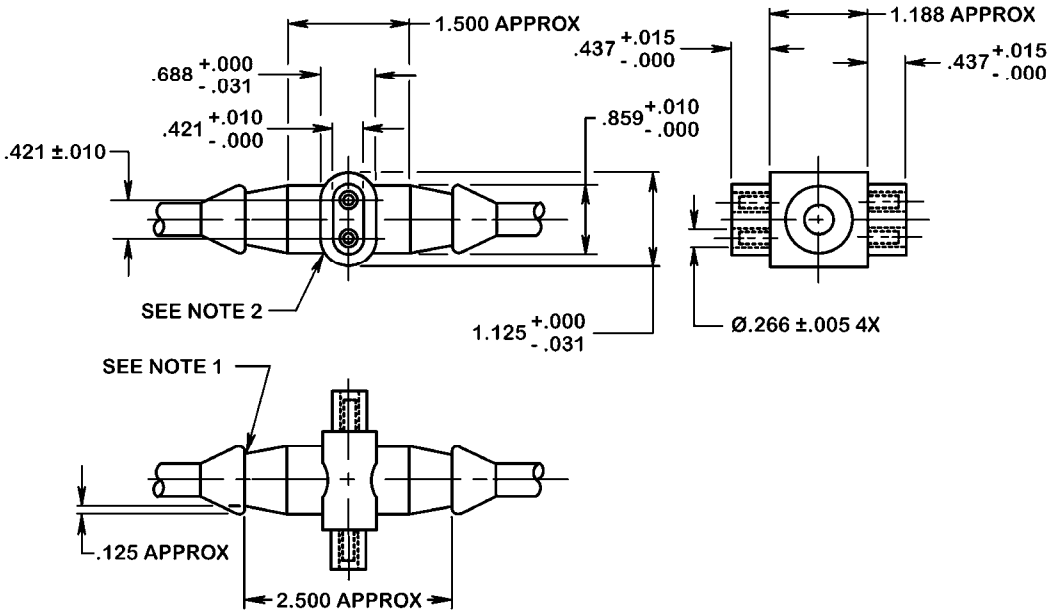
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.



NOTES:
1. This groove to be of sufficient depth, width, or cross section to be used with twine or cord of .125 diameter and need not be of the shape shown.
2. Type I female contacts.
3. Female contacts shall be recessed .125 ±.016 below face.
4. Connector rating: 600 volts, 15 amperes.
5. Voltage drop: Not more than 10 millivolts at 15 amperes.
6. Connecting and disconnecting force: 10-35 pounds.
7. All dimensions are in inches.

FIGURE 3. Plug, molded, (female).

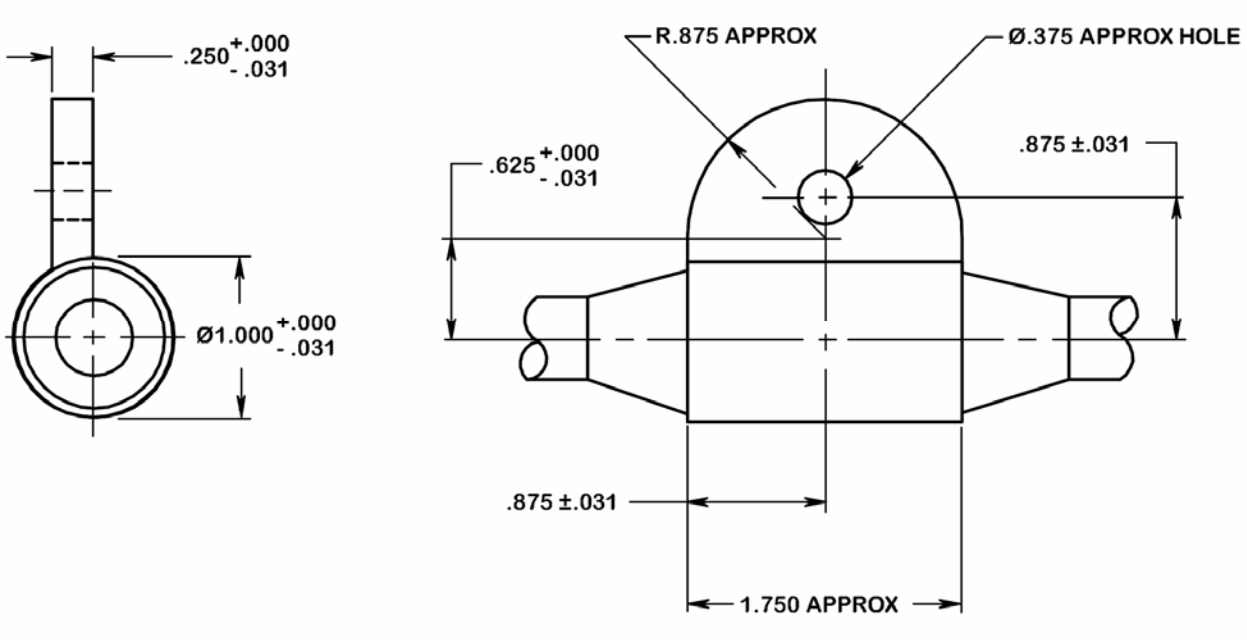
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NOTES:
1. This groove to be of sufficient depth, width, or cross section to be used with twine or cord of .125 diameter and need not be of the shape shown.
2. Type I female contacts.
3. Female contacts shall be recessed .125 ±.016 below face.
4. Connector rating: 600 volts, 15 amperes.
5. Voltage drop: Not more than 10 millivolts at 15 amperes.
6. Connecting and disconnecting force: 10-35 pounds.
7. All dimensions are in inches.

FIGURE 4. Outlet, molded, double, (female).

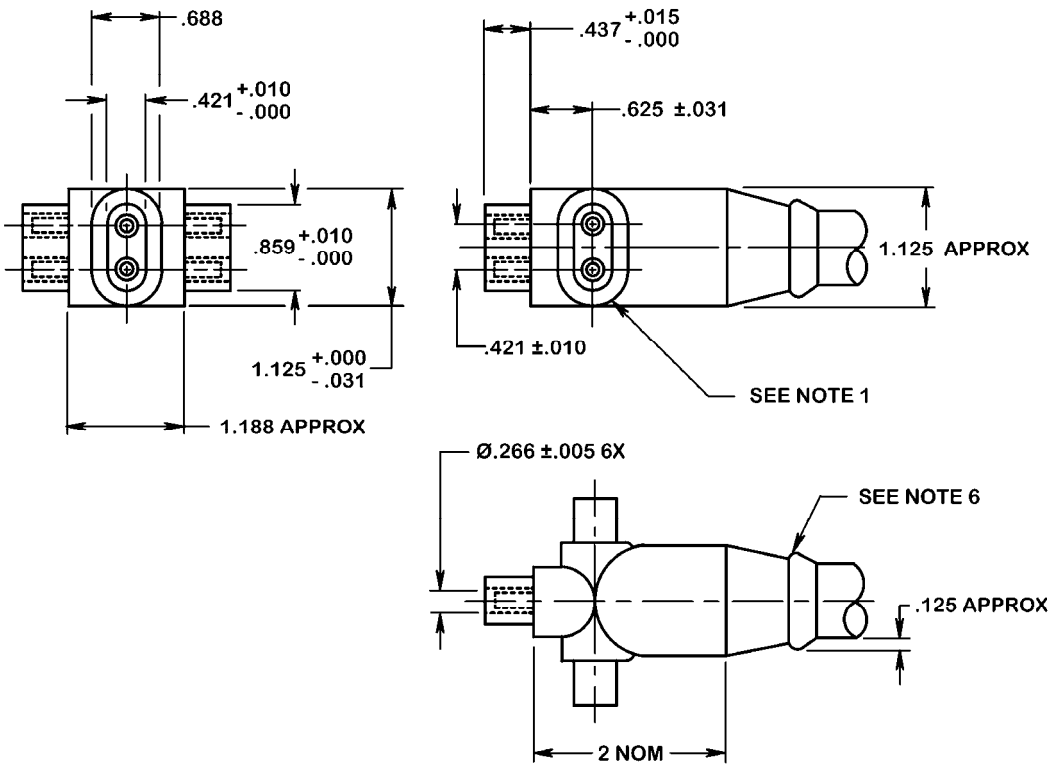
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NOTE:
1. All dimensions are in inches.

FIGURE 5. Hanger, cable - details.

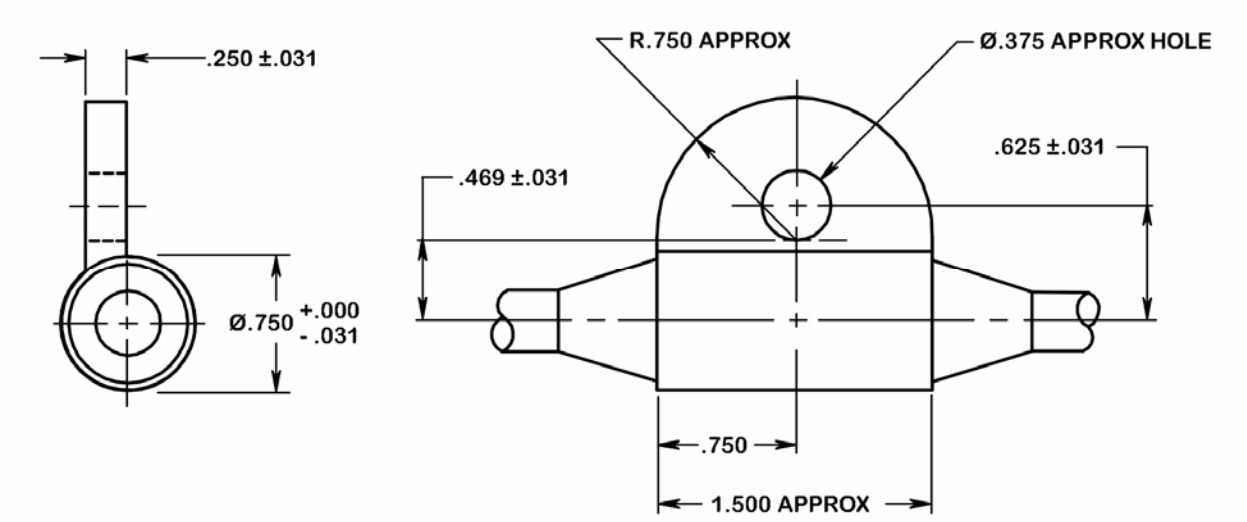
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NOTES:
1. Type I female contacts.
2. Female contacts shall be recessed .125 ±.016 below face.
3. Connector rating: 600 volts, 15 amperes.
4. Voltage drop: Not more than 10 millivolts at 15 amperes.
5. Connecting and disconnecting force: 10-35 pounds.
6. This groove to be of sufficient depth, width, or cross section to be used for securing the dummy shown in fig 8 with twine or cord of .125 diameter and need not be of the shape shown.
7. All dimensions are in inches.

FIGURE 6. Terminal, molded, three-way, (female).

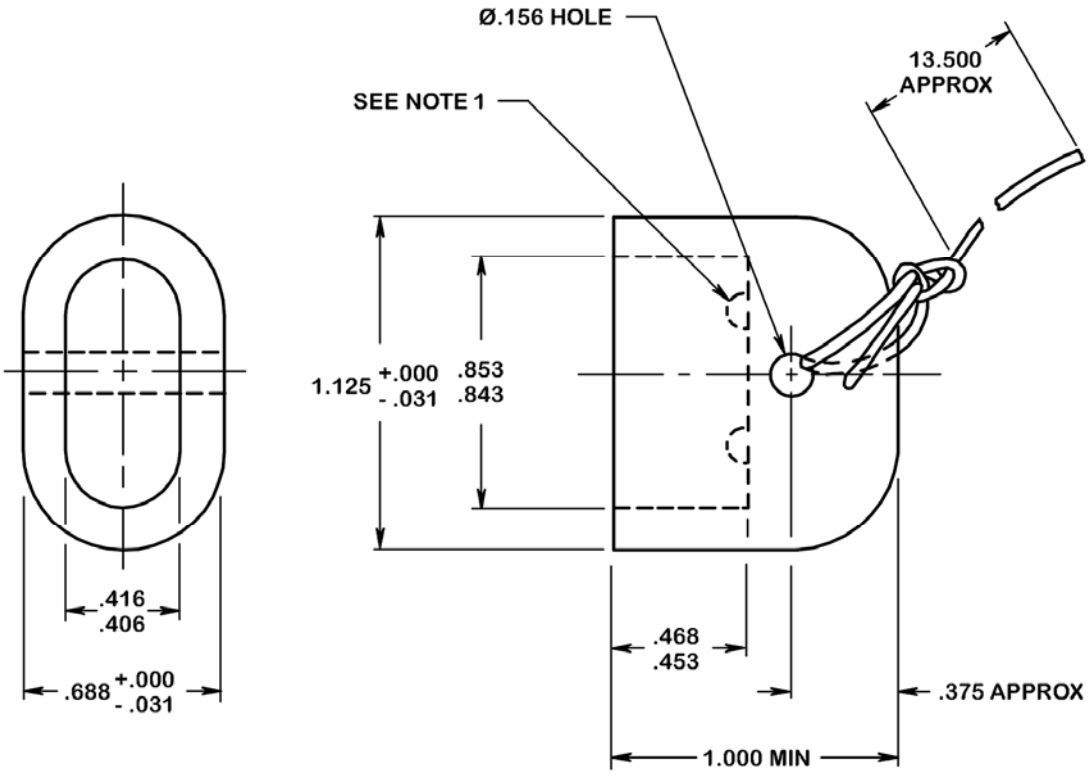
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Check the source to verify that this is the current version before use.



NOTE:
1. All dimensions are in inches.

FIGURE 7. Hanger, cable - details.

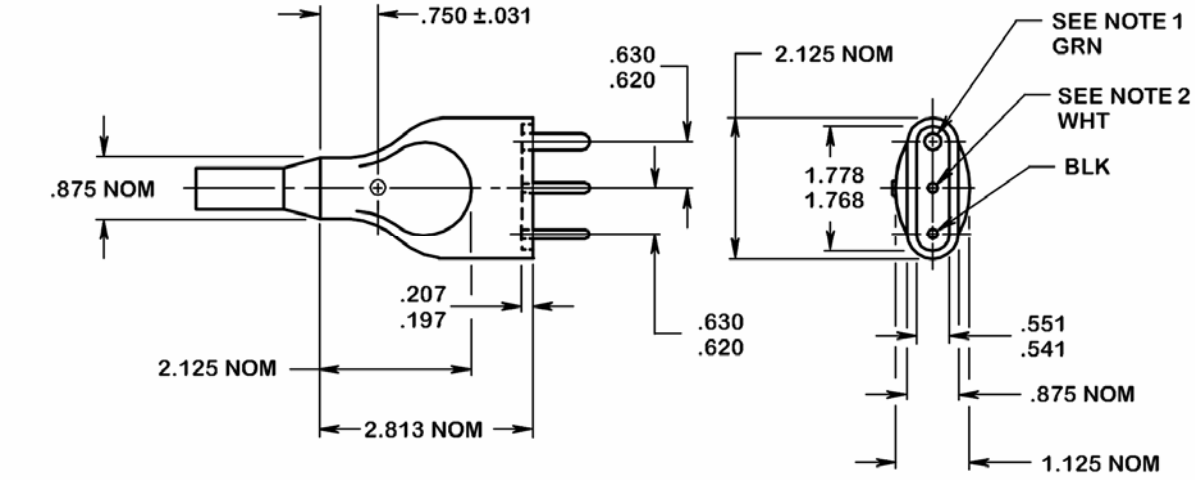
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
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NOTES:
1. Projections may be provided if necessary to insure water tightness.
2. All dimensions are in inches.

FIGURE 8. Dummy with cotton twine, type I, (male).

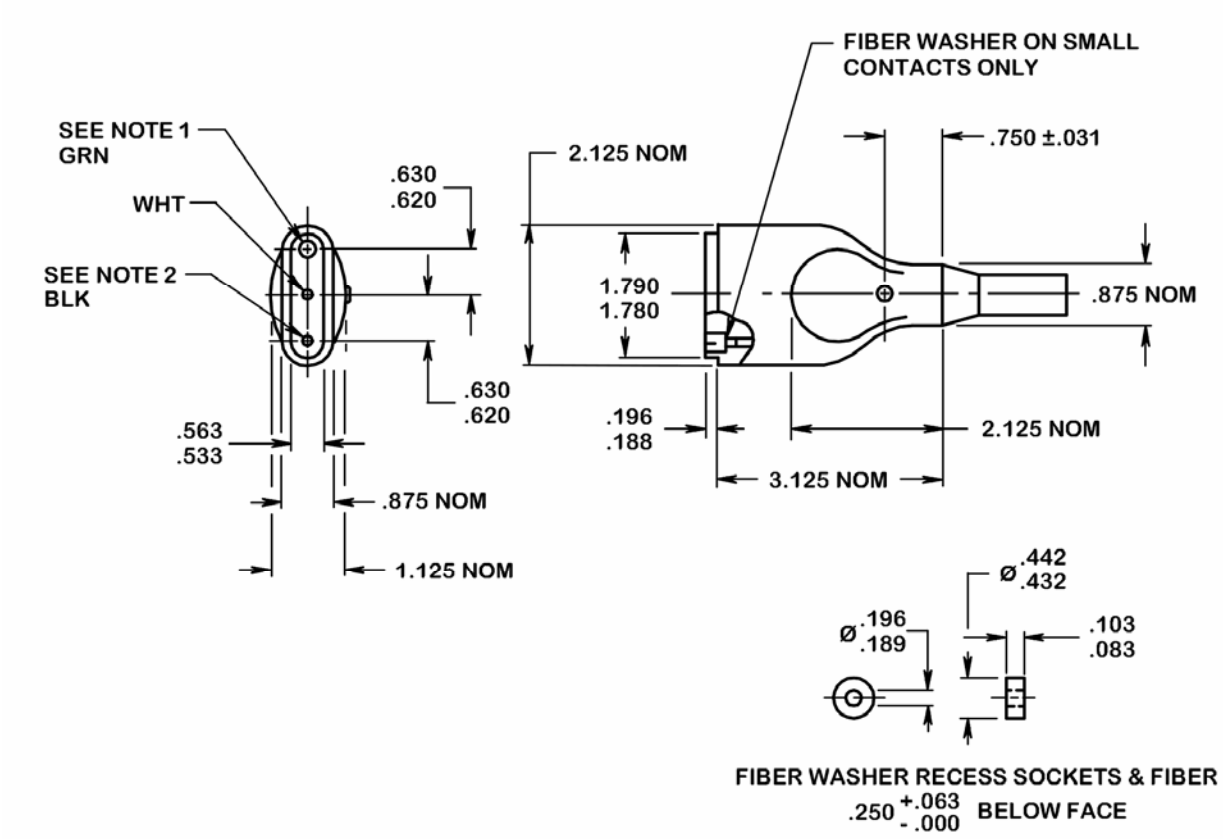
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.



NOTES:
1. Type II polarizing contact.
2. Type II standard contact (2 required).
3. Connector rating: 600 volts, 20 amperes.
4. Voltage drop: Not more than 20 millivolts at 20 amperes.
5. Connecting and disconnecting force: 10-35 pounds.
6. All dimensions are in inches.

FIGURE 9. Plug, 3-pole, (male).

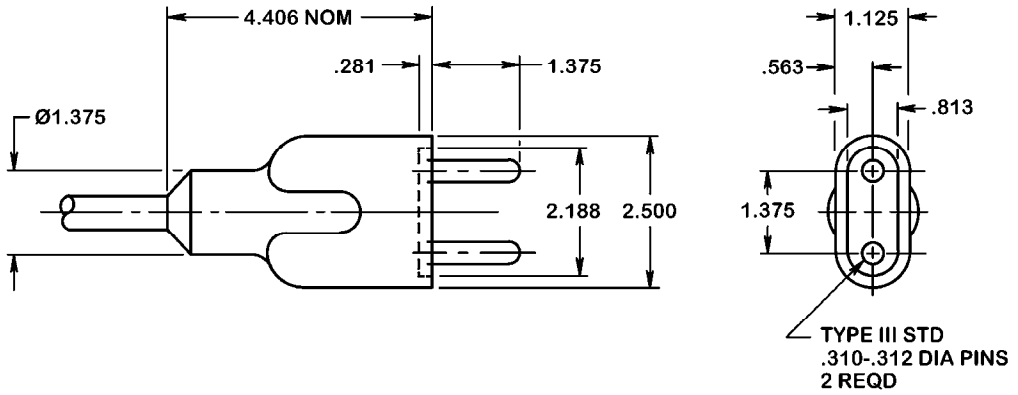
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NOTES:
1. Type II polarizing contact.
2. Type II standard contact (2 required).
3. Connector rating: 600 volts, 20 amperes.
4. Voltage drop: Not more than 20 millivolts at 20 amperes.
5. Connecting and disconnecting force: 10-35 pounds.
6. All dimensions are in inches.

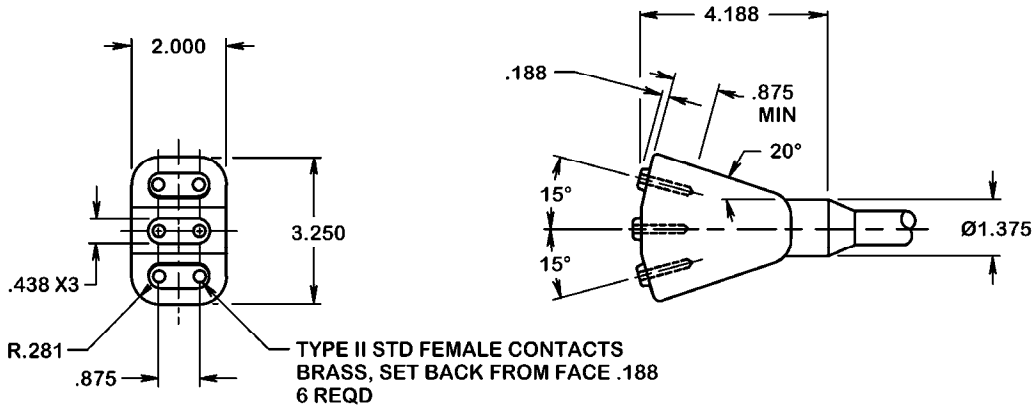
FIGURE 10. Plug, 3-pole, (female).

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Check the source to verify that this is the current version before use.



NOTES:
1. Connector rating: 600 volts, 25 amperes.
2. Voltage drop: Not more than 20 millivolts at 25 amperes.
3. Connecting and disconnecting force: 15-35 pounds.
4. All dimensions are in inches.

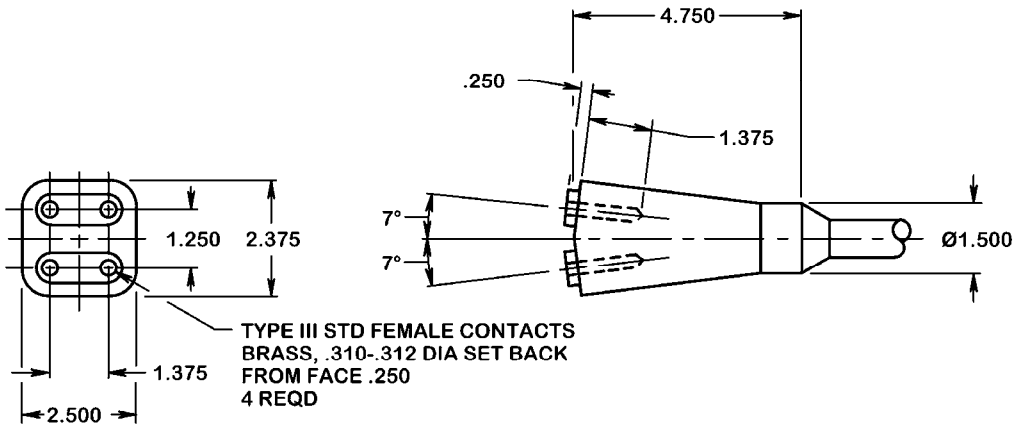
FIGURE 11. Connector, plug, electrical, waterproof, (male).



NOTES:
1. Connector rating: 600 volts, 15 amperes.
2. Voltage drop: Not more than 10 millivolts at 15 amperes.
3. Connecting and disconnecting force: 10-35 pounds.
4. All dimensions are in inches.

FIGURE 12. Connector, waterproof, (female).

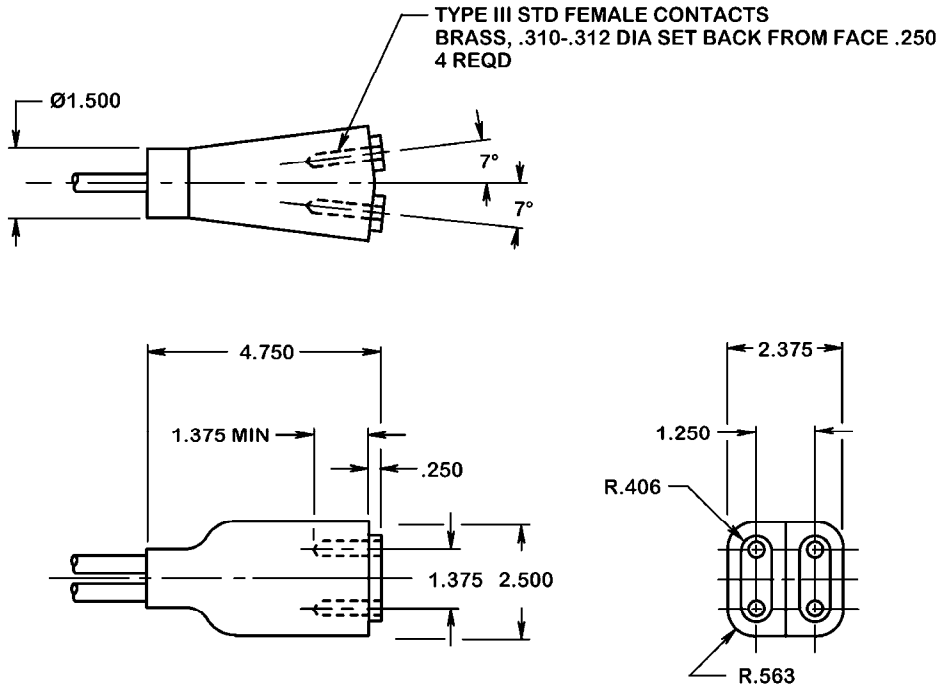
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Check the source to verify that this is the current version before use.



NOTES:
1. Connector rating: 600 volts, 20 amperes.
2. Voltage drop: Not more than 10 millivolts at 20 amperes.
3. Connecting and disconnecting force: 10-35 pounds.
4. All dimensions are in inches.

FIGURE 13. Connector, plug, floodlight, waterproof, (female).

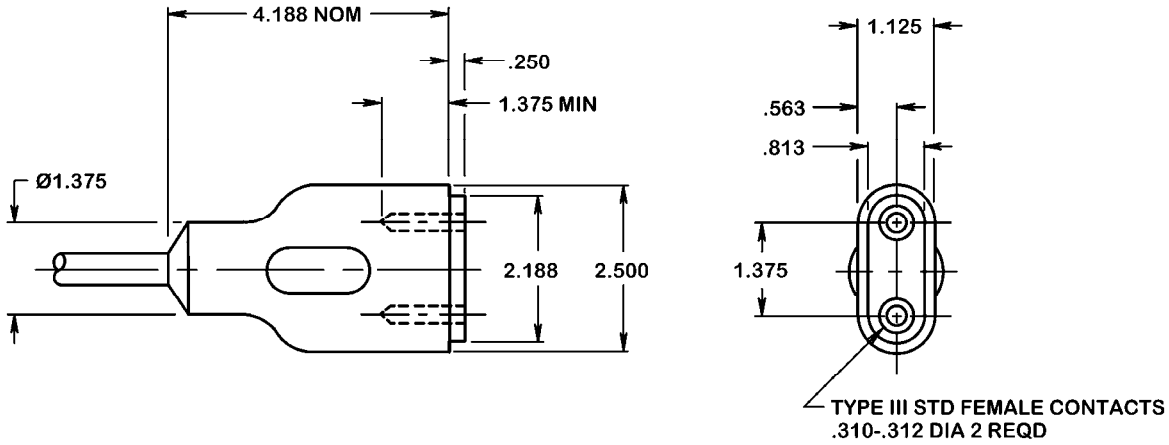
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Check the source to verify that this is the current version before use.



NOTES:
1. Connector rating: 600 volts, 20 amperes.
2. Voltage drop: Not more than 20 millivolts at 20 amperes.
3. Connecting and disconnecting force: 15-35 pounds.
4. All dimensions are in inches.

FIGURE 14. Connector, waterproof, (female).

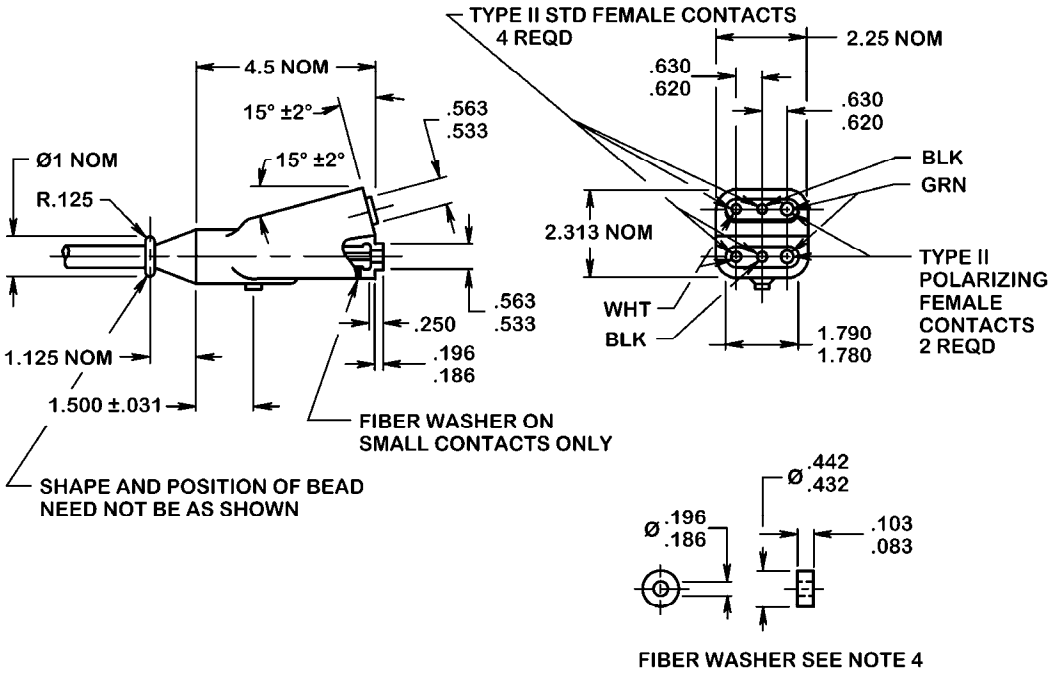
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Check the source to verify that this is the current version before use.



NOTES:
1. Connector rating: 600 volts, 25 amperes.
2. Voltage drop: Not more than 20 millivolts at 25 amperes.
3. Connecting and disconnecting force: 15-35 pounds.
4. All dimensions are in inches.

FIGURE 15. Connector, plug, electrical, waterproof, (female).

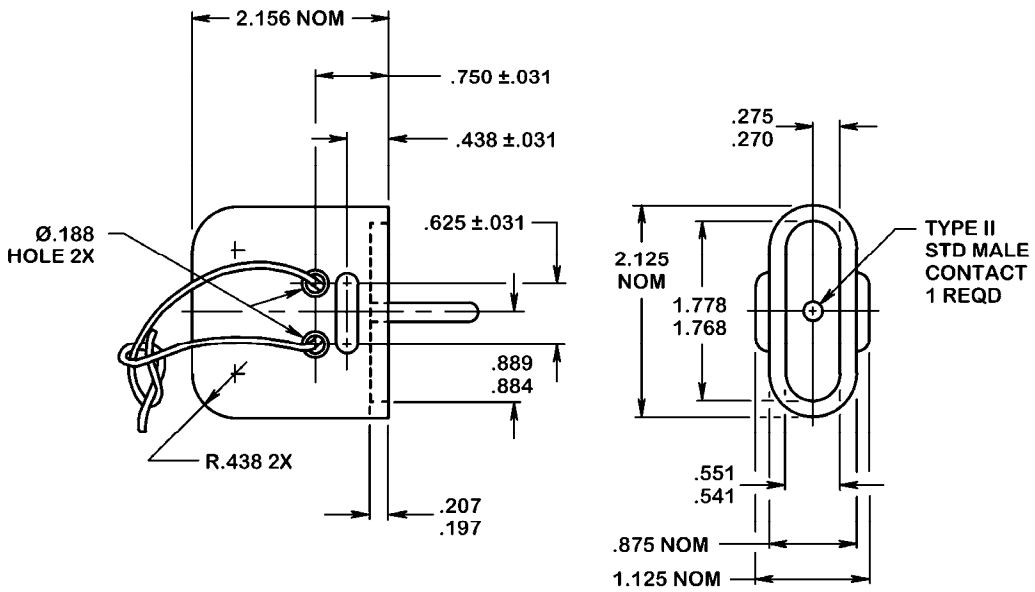
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.



NOTES:
1. Connector rating: 600 volts, 20 amperes.
2. Voltage drop: Not more than 10 millivolts at 20 amperes.
3. Connecting and disconnecting force: 10-35 pounds.
4. Recess sockets and fiber washers .250 +.063/-.000 below face.
5. All dimensions are in inches.

FIGURE 16. Connector, two-way, three-conductor, (female).

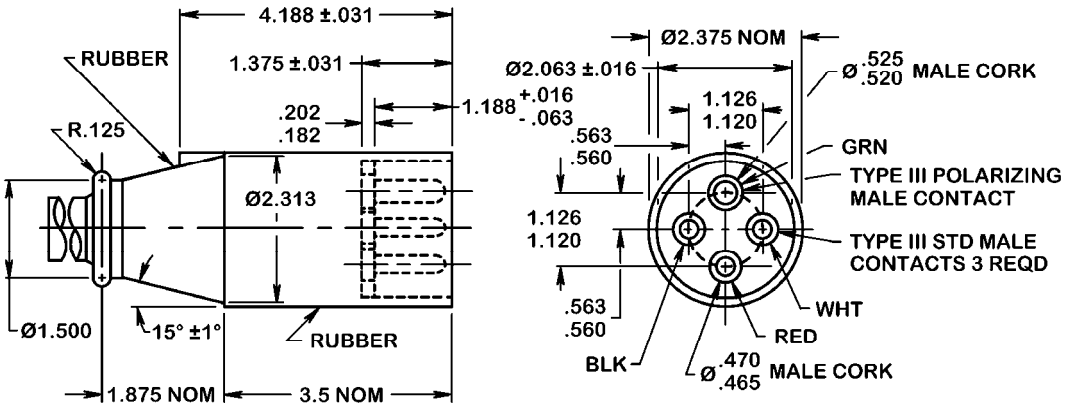
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
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NOTES:
1. Seal shall be equipped with 12 in. cotton twine, .125 diameter.
2. All dimensions are in inches.

FIGURE 17. Dummy, (male).

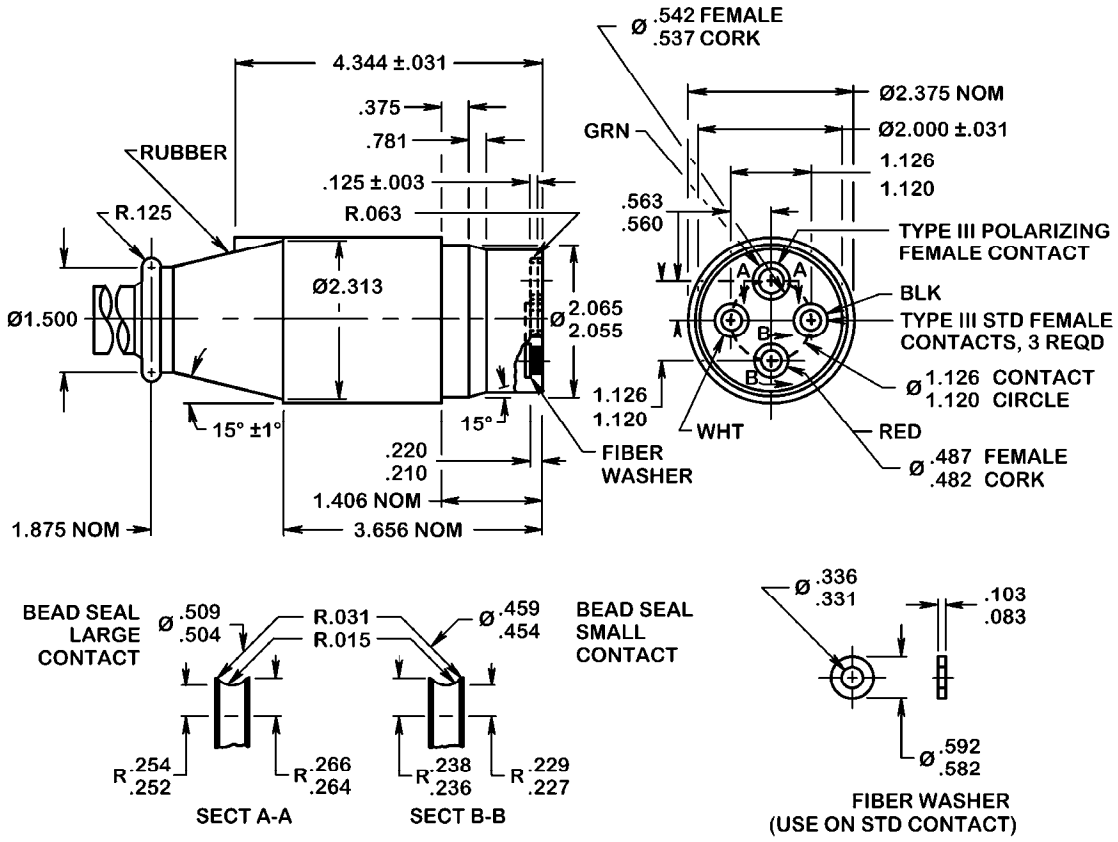
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NOTES:
1. Connector rating: 600 volts, 65 amperes.
2. Voltage drop: Not more than 15 millivolts at 65 amperes.
3. Connecting and disconnecting force: 40-100 pounds.
4. All dimensions are in inches.

FIGURE 18. Plug, molded, (male).

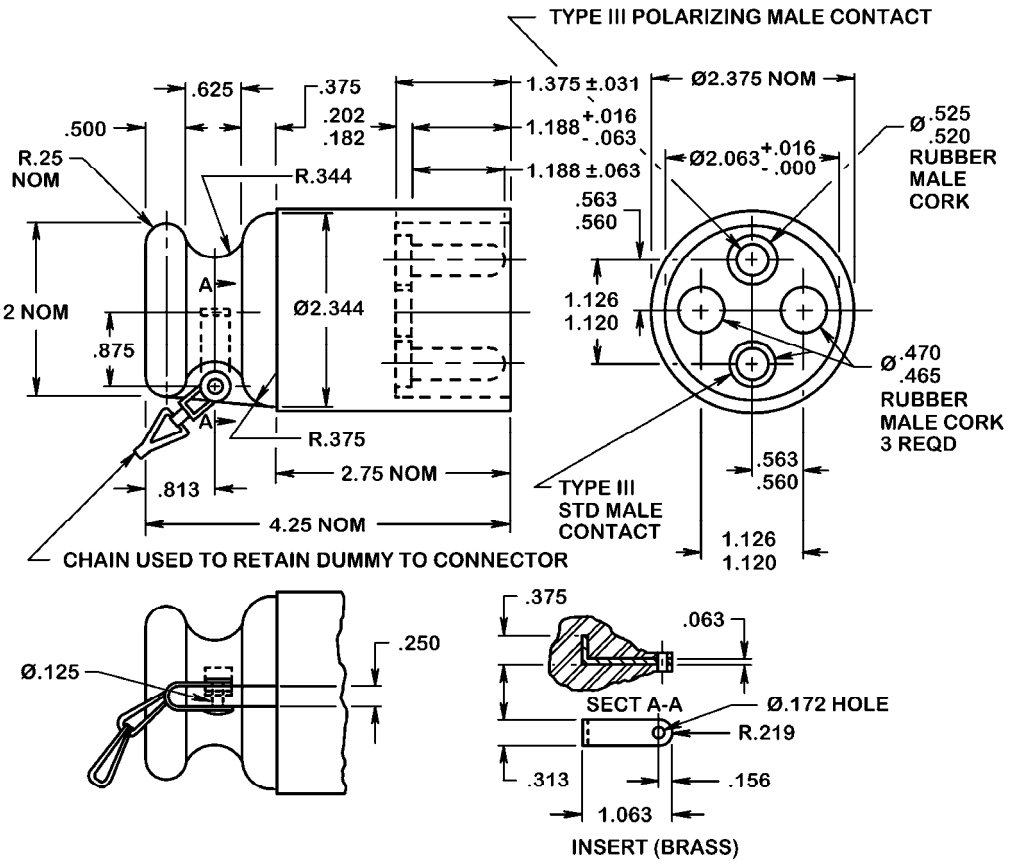
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.



NOTES:
1. Connector rating: 600 volts, 65 amperes.
2. Voltage drop: Not more than 15 millivolts at 65 amperes.
3. Connecting and disconnecting force: 40-100 pounds.
4. All dimensions are in inches.

FIGURE 19. Connector, plug, (female).

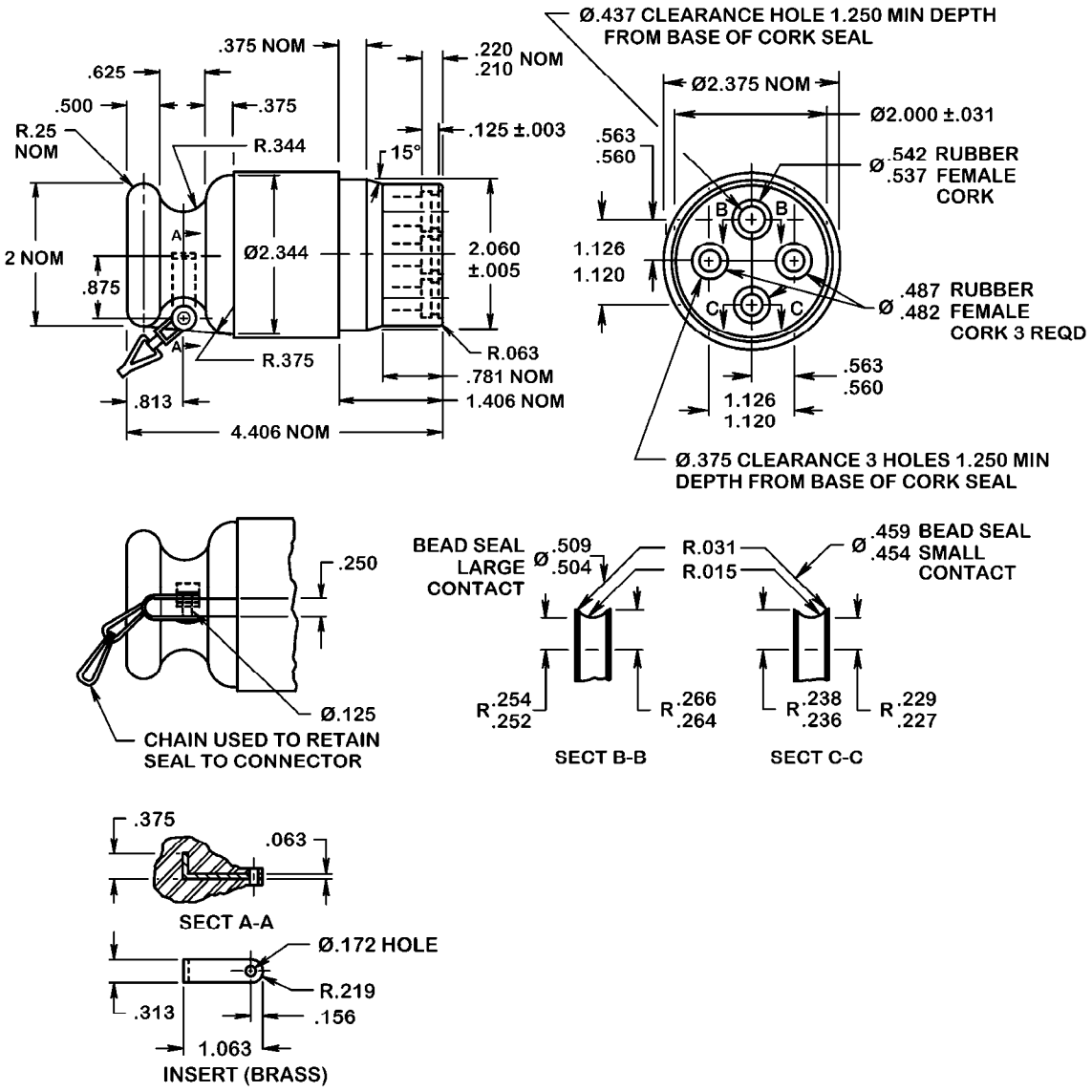
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NOTE:
1. All dimensions are in inches.

FIGURE 20. Dummy, molded.

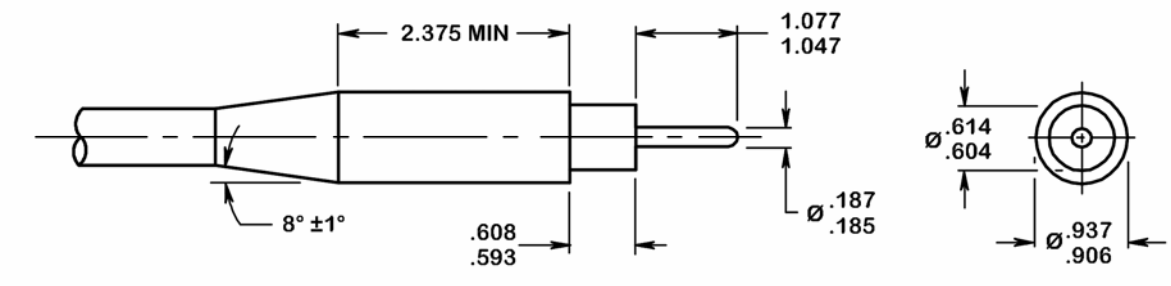
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
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NOTE:
1. All dimensions are in inches.

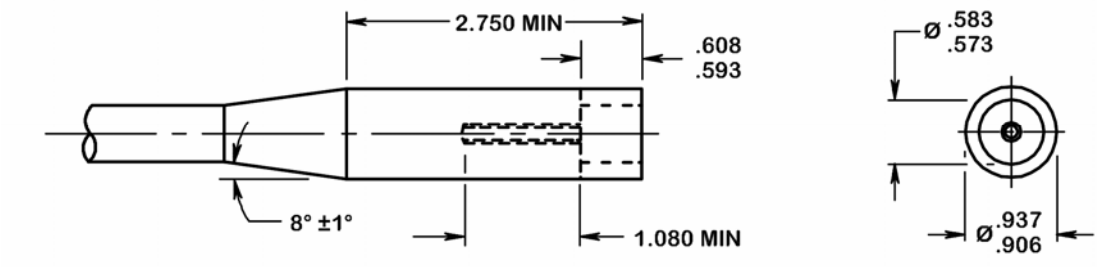
FIGURE 21. Dummy, molded, type II, (female).

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NOTES:
1. Rating: 25 amperes, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. All dimensions are in inches.

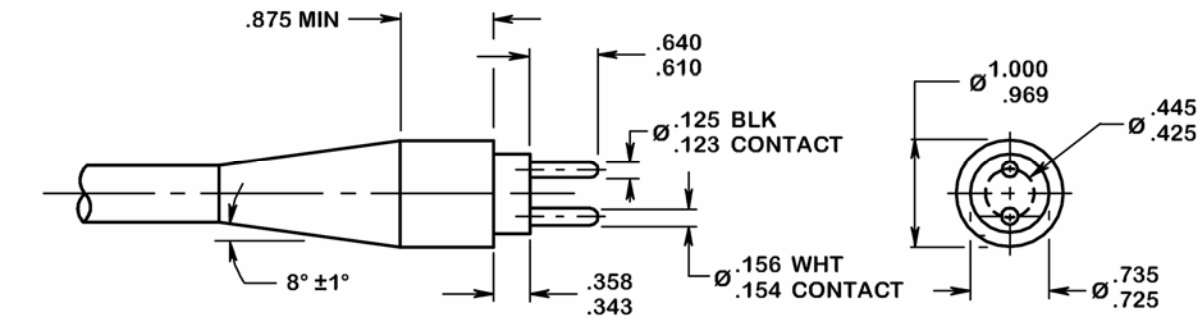
FIGURE 22. Connector, plug, single-pole, (male).



NOTES:
1. Rating: 25 amperes, 1500 volts to ground.
2. Socket shall be recessed .125 max.
3. Socket ID: .186–.189 before splitting.
4. Connecting and disconnecting force: 10-35 pounds.
5. All dimensions are in inches.

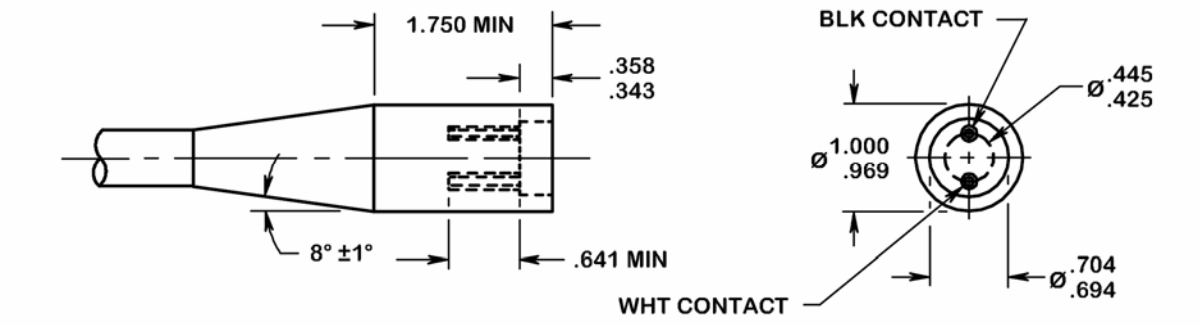
FIGURE 23. Connector, plug, single-pole, (female).

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NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. All dimensions are in inches.

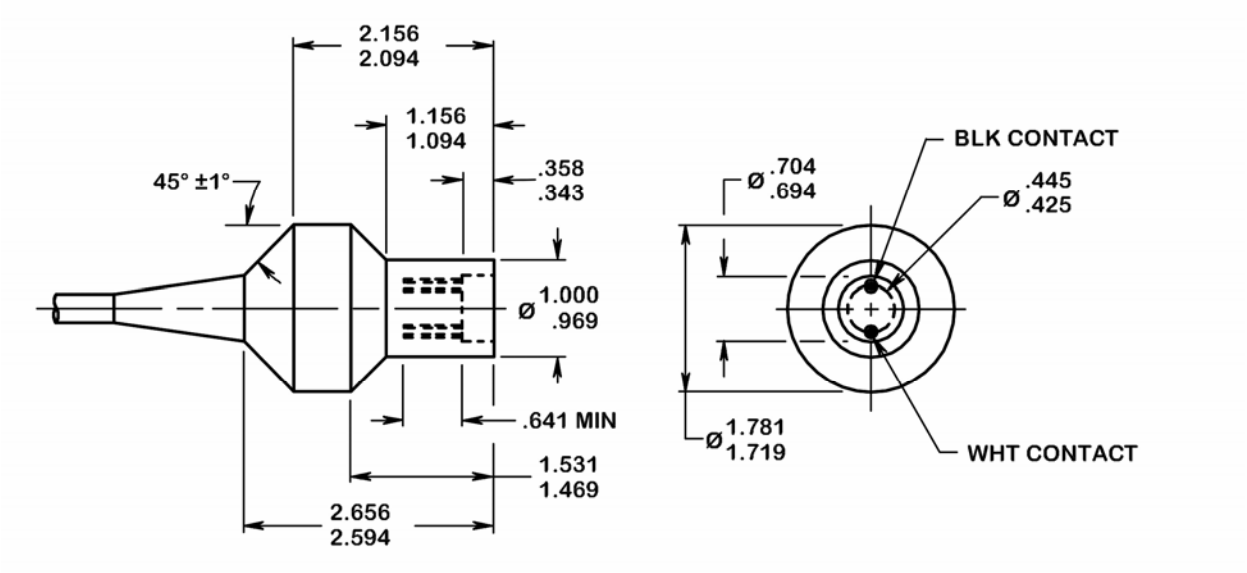
FIGURE 24. Connector, plug, 2-pole, (male).



NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Socket shall be recessed .125 max.
3. Before splitting, large socket ID .156–.158 and small socket ID .125–.127.
4. Connecting and disconnecting force: 10-35 pounds.
5. All dimensions are in inches.

FIGURE 25. Connector, plug, 2-pole, (female).

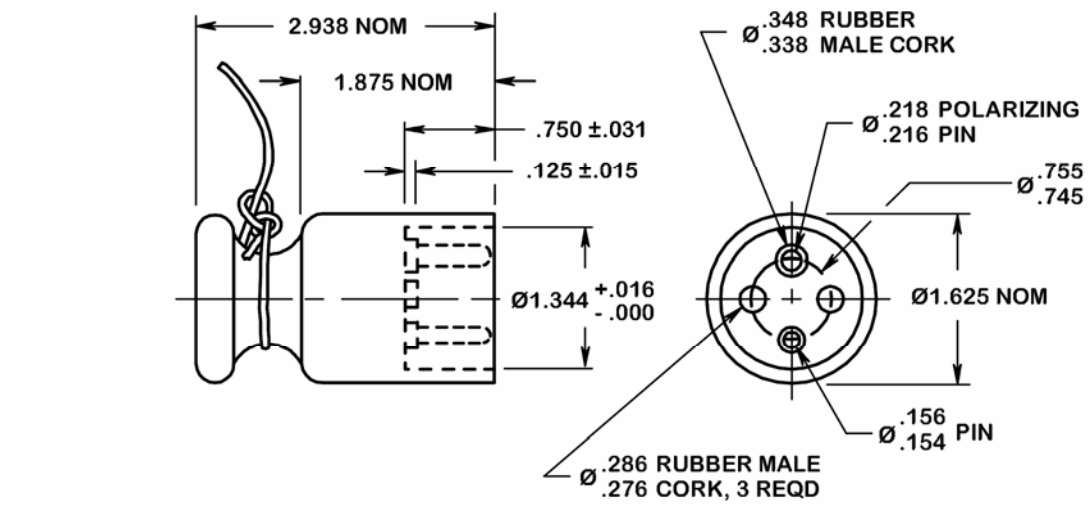
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.



NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Socket shall be recessed .125 max.
3. Before splitting, large socket ID .156–.158 and small socket ID .125–.127.
4. All dimensions are in inches.

FIGURE 26. Connector, plug, 2-pole, (female).

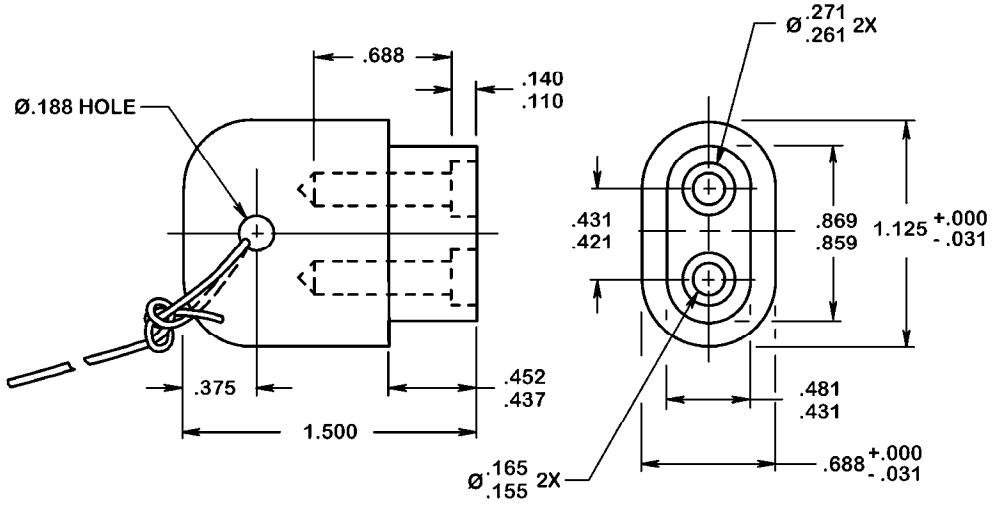
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
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NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. Dummy plug shall be equipped with cotton twine 14.0 in. long.
4. All dimensions are in inches.

FIGURE 27. Connector, plug, dummy, (male).

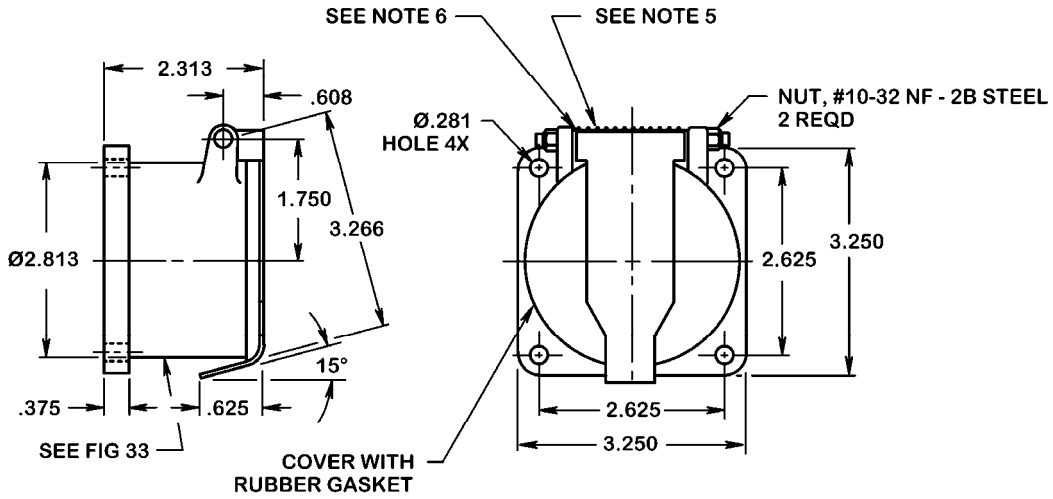
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.



NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. Dummy plug shall be equipped with cotton twine 14 in. long.
4. All dimensions are in inches.

FIGURE 28. Connector, plug, dummy, (female).

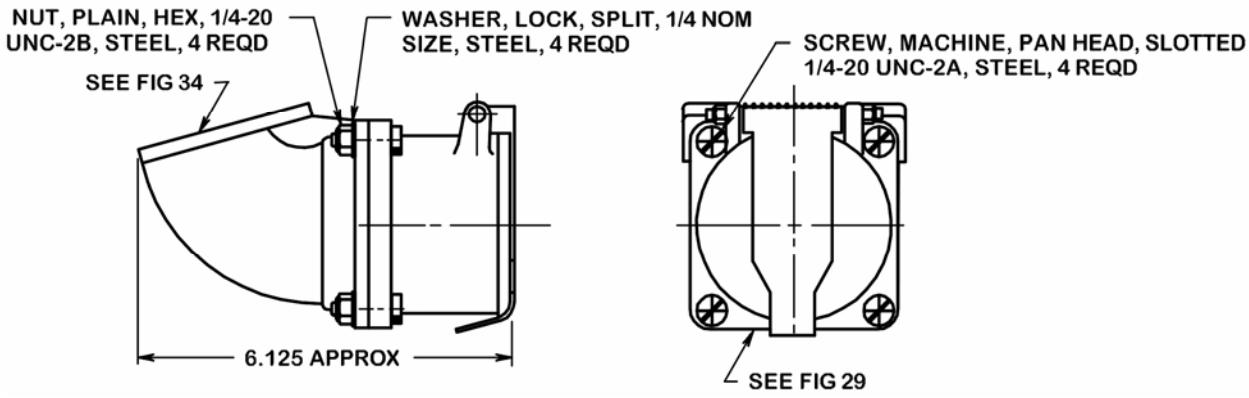
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
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NOTES:
1. Rating: 600 volts, 65 amperes.
2. Connecting and disconnecting force: 40-100 pounds.
3. Cadmium or zinc plate all steel parts.
4. Used with inserts, figure 35 and figure 43.
5. Spring data:
   a. Material: Steel
   b. Wire diameter: .090
   c. Inside diameter: .281
   d. Free length: 1.188
   e. Number of coils: 11.875
6. Pin data:
   a. Material: Steel
   b. Length: 2.750
   c. Thread: #10-32 NF-2A (both ends)
7. All dimensions are in inches.

FIGURE 29. Shell, connector, electrical.

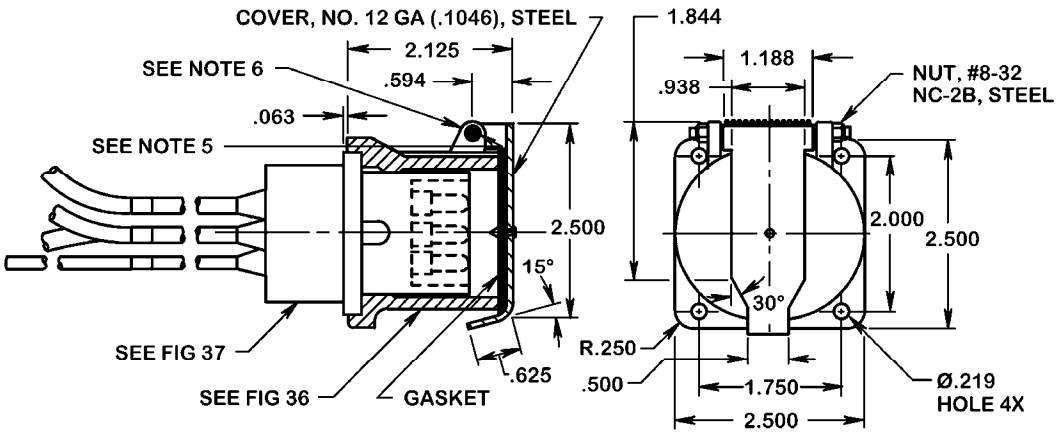
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NOTES:
1. Cadmium or zinc plate all steel parts.
2. Used with insert, figure 43.
3. All dimensions are in inches.

FIGURE 30. Shell, connector, electrical.

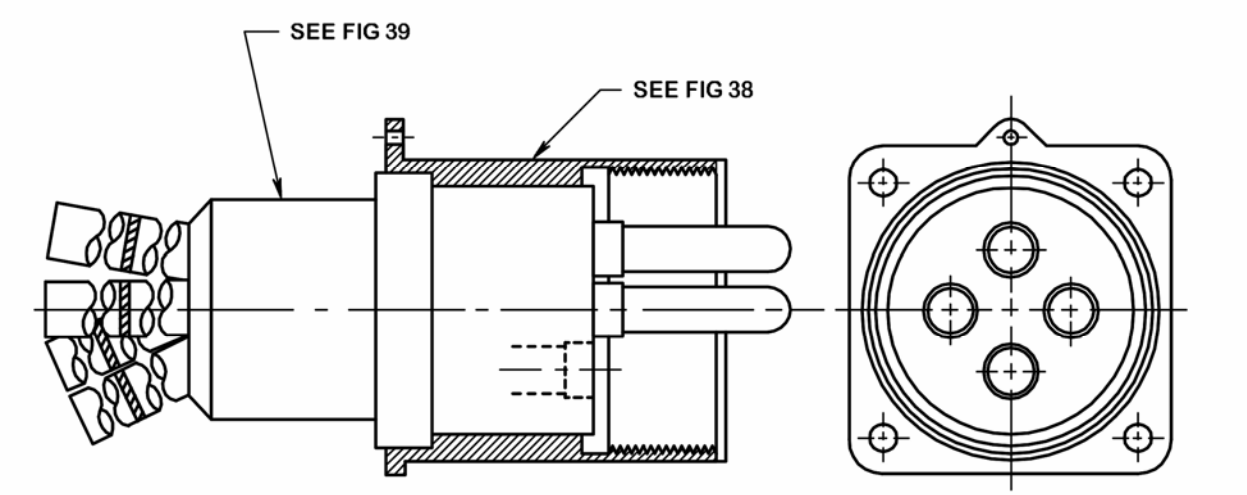
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NOTES:
1. Rating: 600 volts, 20 amperes.
2. Connecting and disconnecting force: 10-35 pounds.
3. Cadmium or zinc plate all steel parts.
4. Used with inserts, figure 35 and figure 43.
5. Spring data:
   a. Material: Steel
   b. Wire diameter: .063
   c. Outside diameter: .366
   d. Free length: 1.156
   e. Number of coils: 13.750
6. Pin data:
   a. Material: Steel
   b. Dia: .156
   c. Length: 2.281
   d. Thread: #8-32 NC-2A (both ends)
7. All dimensions are in inches.

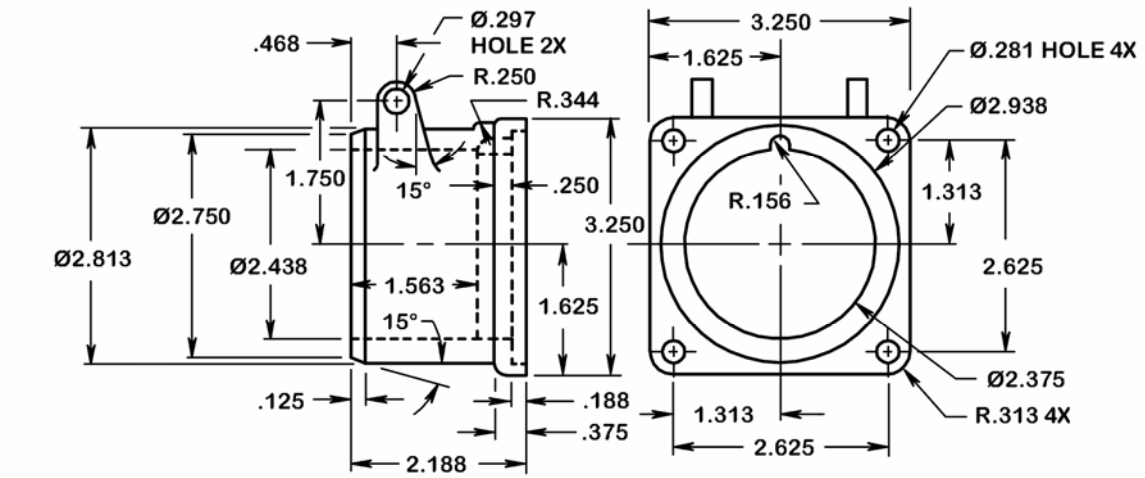
FIGURE 31. Receptacle, connector, electrical.

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NOTES:
1. Rating: 600 volts, 230 amperes.
2. Connecting and disconnecting force: 15-35 pounds.

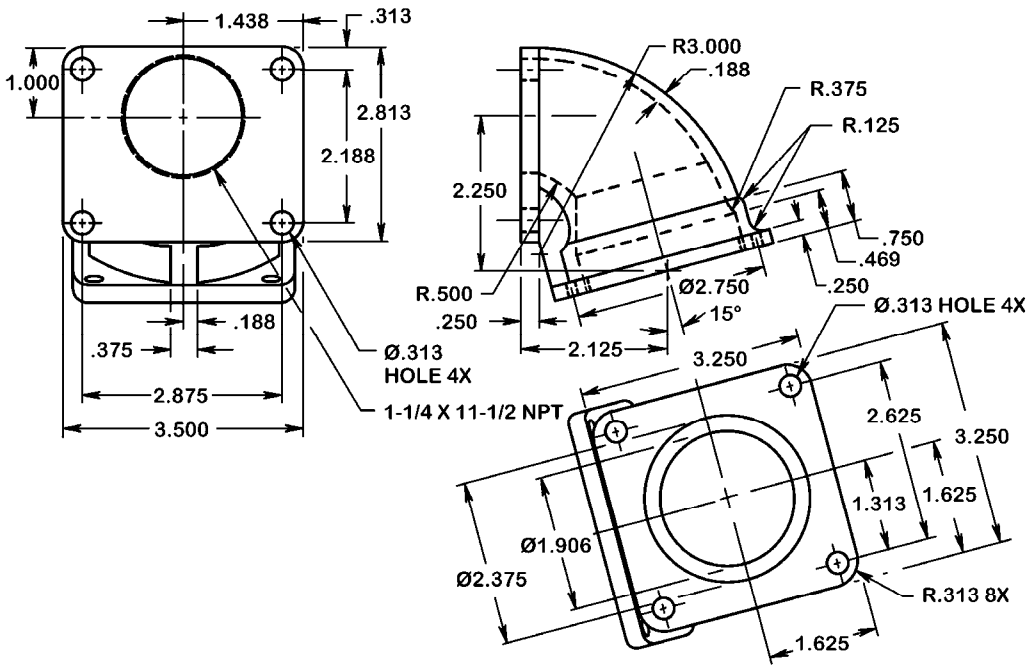
FIGURE 32. Receptacle, connector, electrical.



NOTE:
1. All dimensions are in inches.

FIGURE 33. Shell, connector, electrical.

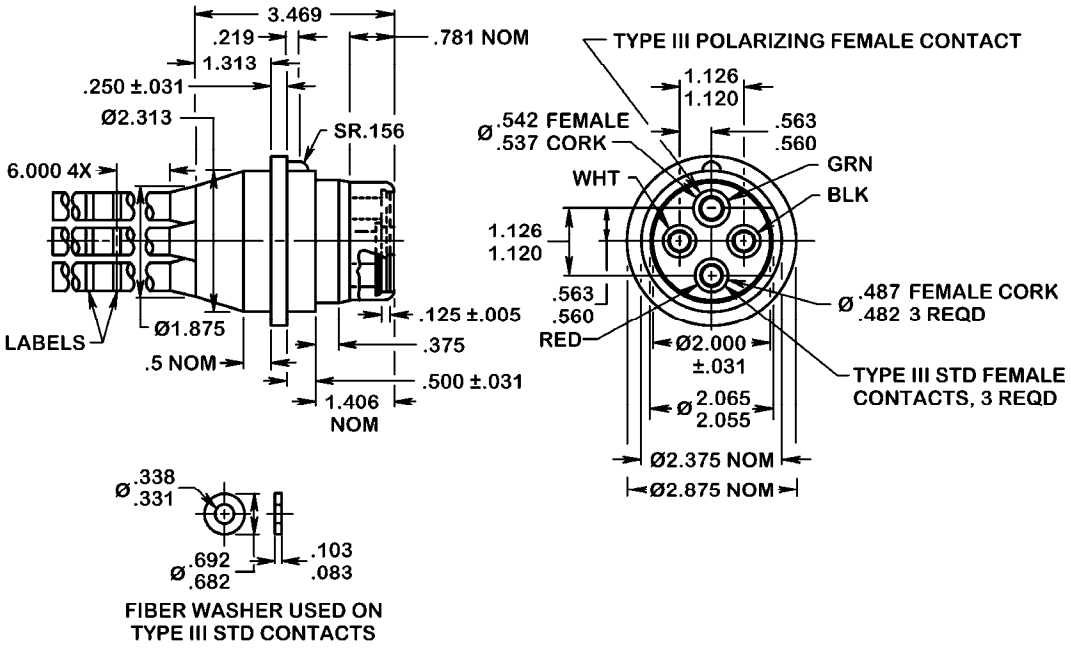
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NOTE:
1. All dimensions are in inches.

FIGURE 34. Shell, connector, electrical.

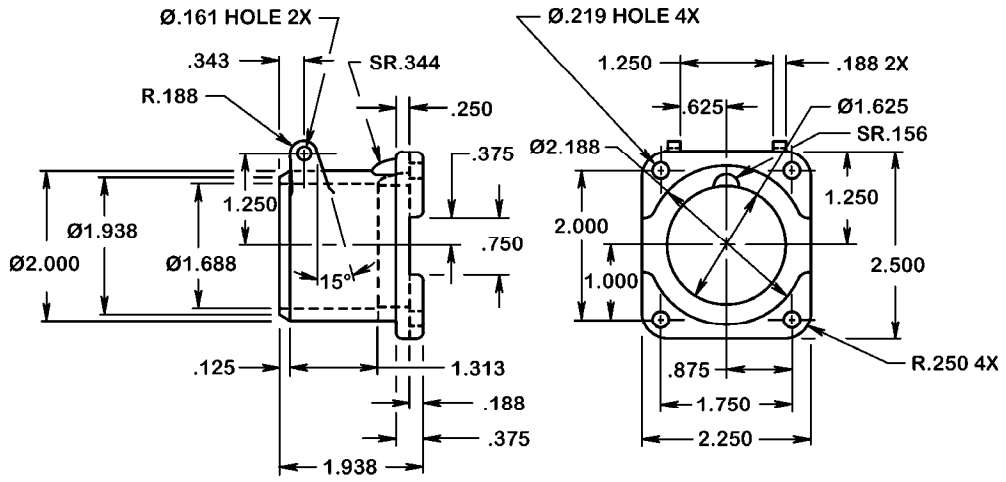
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NOTE:
1. All dimensions are in inches.

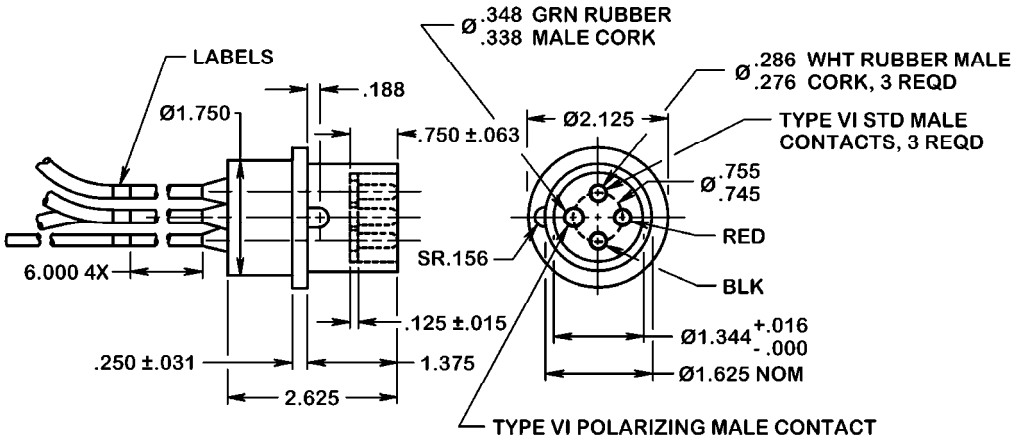
FIGURE 35. Insert, connector, electrical, (female).

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NOTE:
1. All dimensions are in inches.

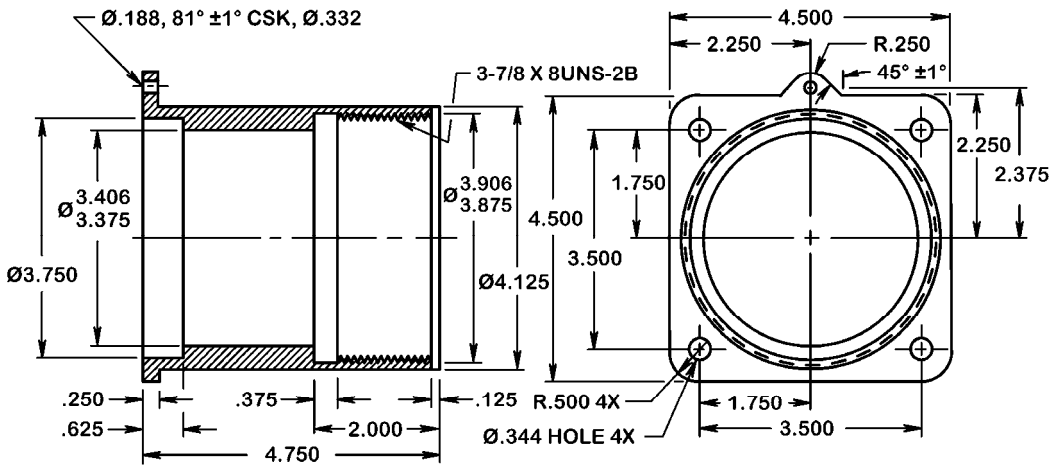
FIGURE 36. Shell, connector, electrical.



NOTE:
1. All dimensions are in inches.

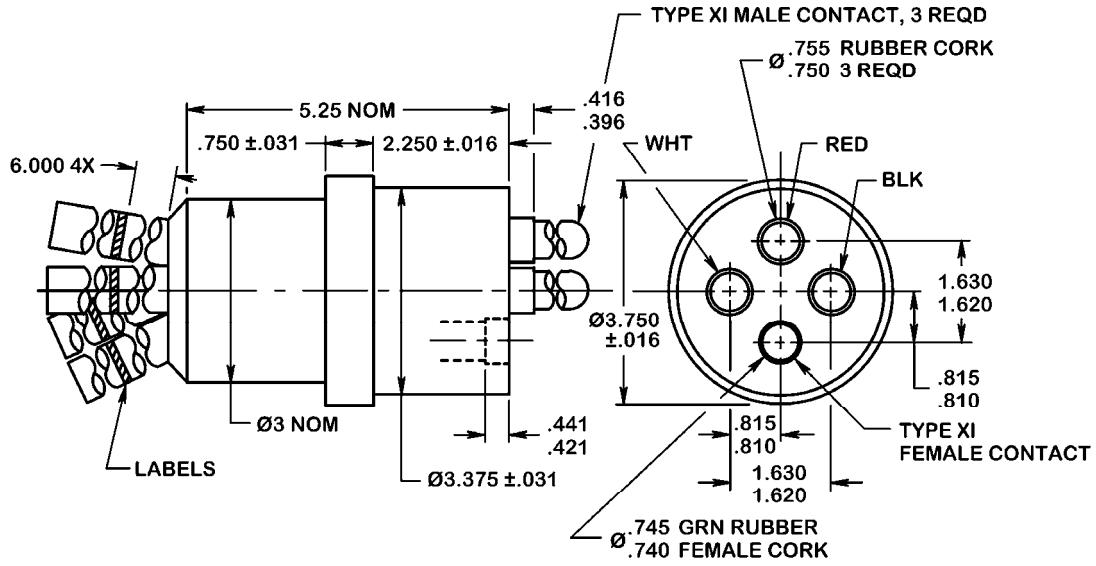
FIGURE 37. Insert, connector, electrical, (male).

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NOTE:
1. Used with insert figure 39.
2. All dimensions are in inches.

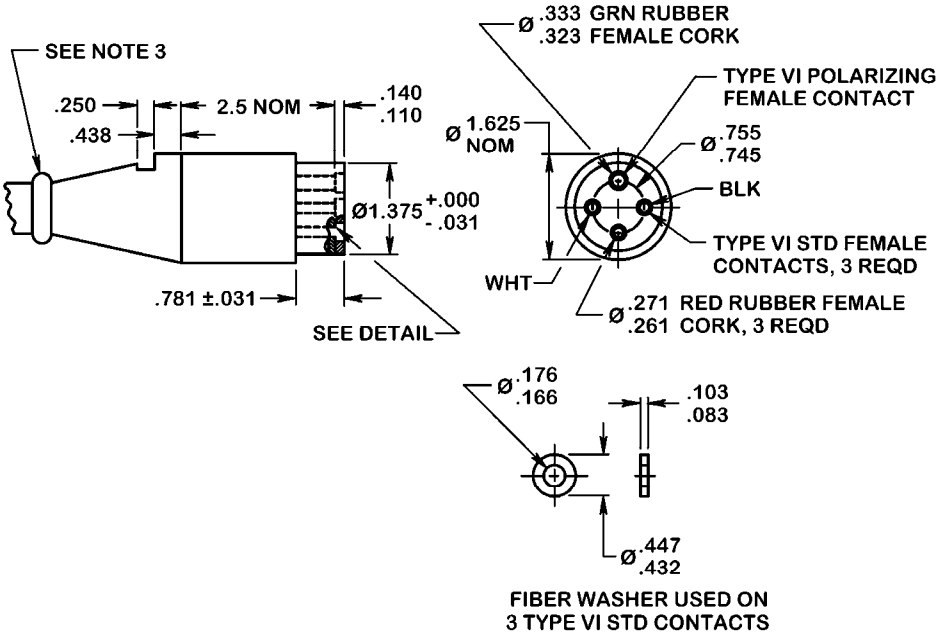
FIGURE 38. Shell, connector, electrical.



NOTE:
1. All dimensions are in inches.

FIGURE 39. Insert, connector, electrical, (male).

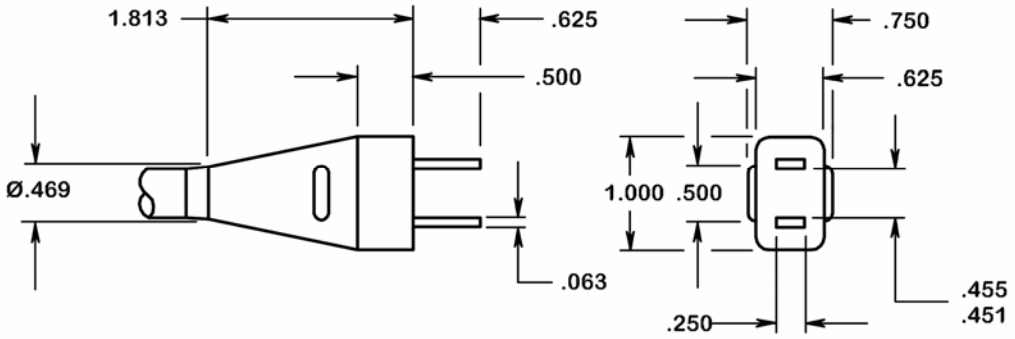
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NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. Dimensions of groove to be sufficient to retain a .125 diameter cord or twine tied on to the connector. Shape and position need not be as shown.
4. All dimensions are in inches.

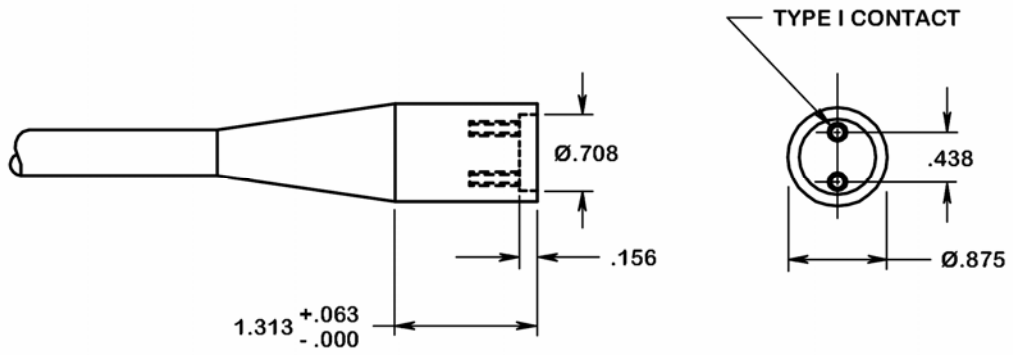
FIGURE 40. Connector, plug, electrical, (female).

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NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. All dimensions are in inches.

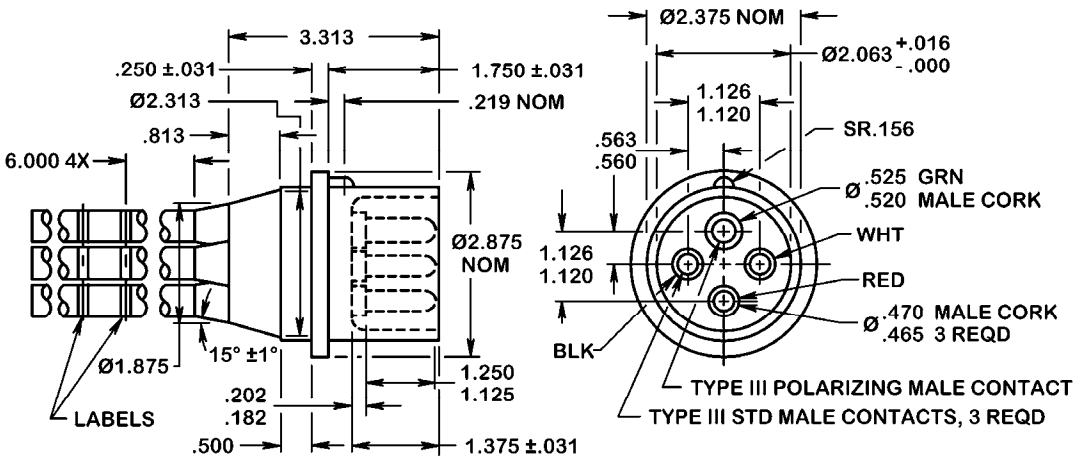
FIGURE 41. Connector, plug, 2-pole, (male).



NOTES:
1. Rating: 20 amperes, 600 volts between contacts, 1500 volts to ground.
2. Connecting and disconnecting force: 10-35 pounds.
3. All dimensions are in inches.

FIGURE 42. Connector, plug, 2-pole, (female).

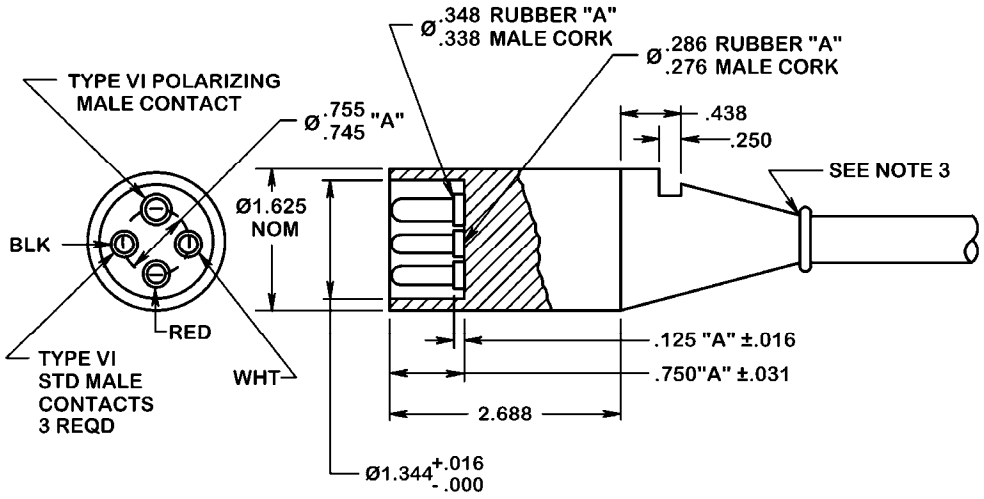
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NOTE:
1. All dimensions are in inches.

FIGURE 43. Insert, connector, electrical, (male).

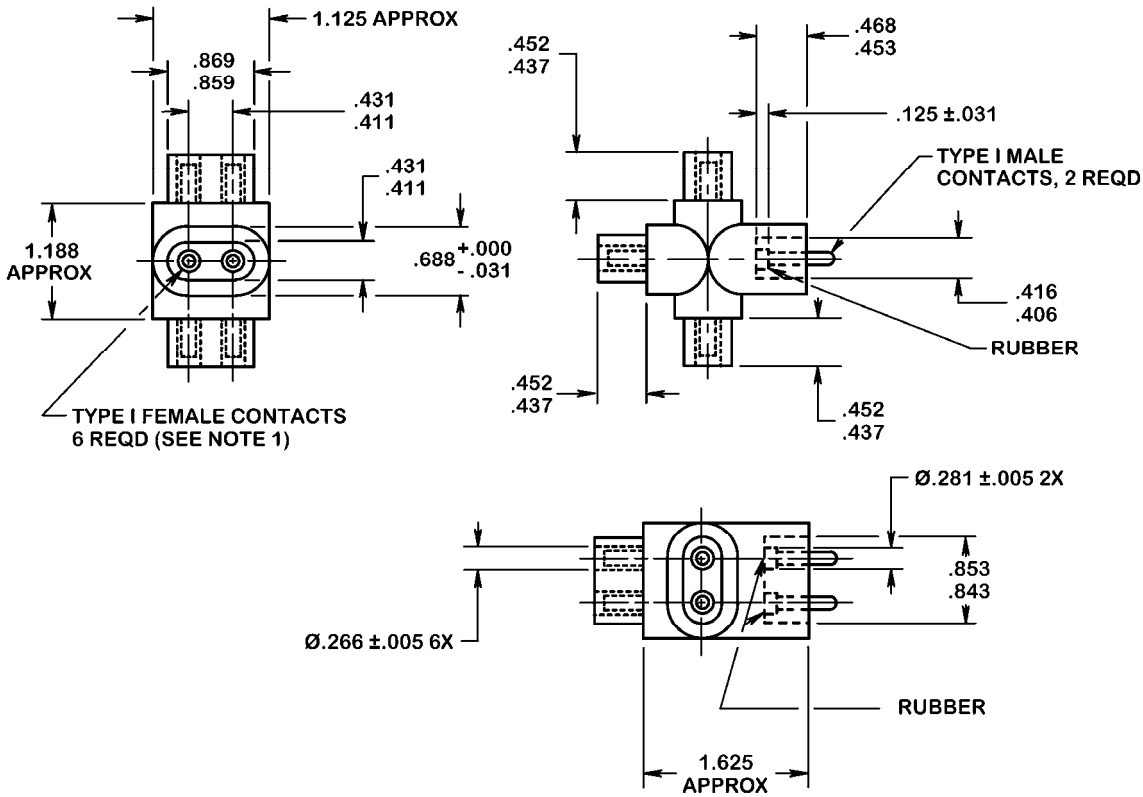
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NOTES:
1. Rating: 600 volts, 20 amperes.
2. Connecting and disconnecting force: 10-35 pounds.
3. Bead to be of sufficient dimension to retain a .125 diameter cord or twine tied-on to the connector. Shape and position need not be as shown.
4. All dimensions are in inches.

FIGURE 44. Connector, plug, watertight, (male).

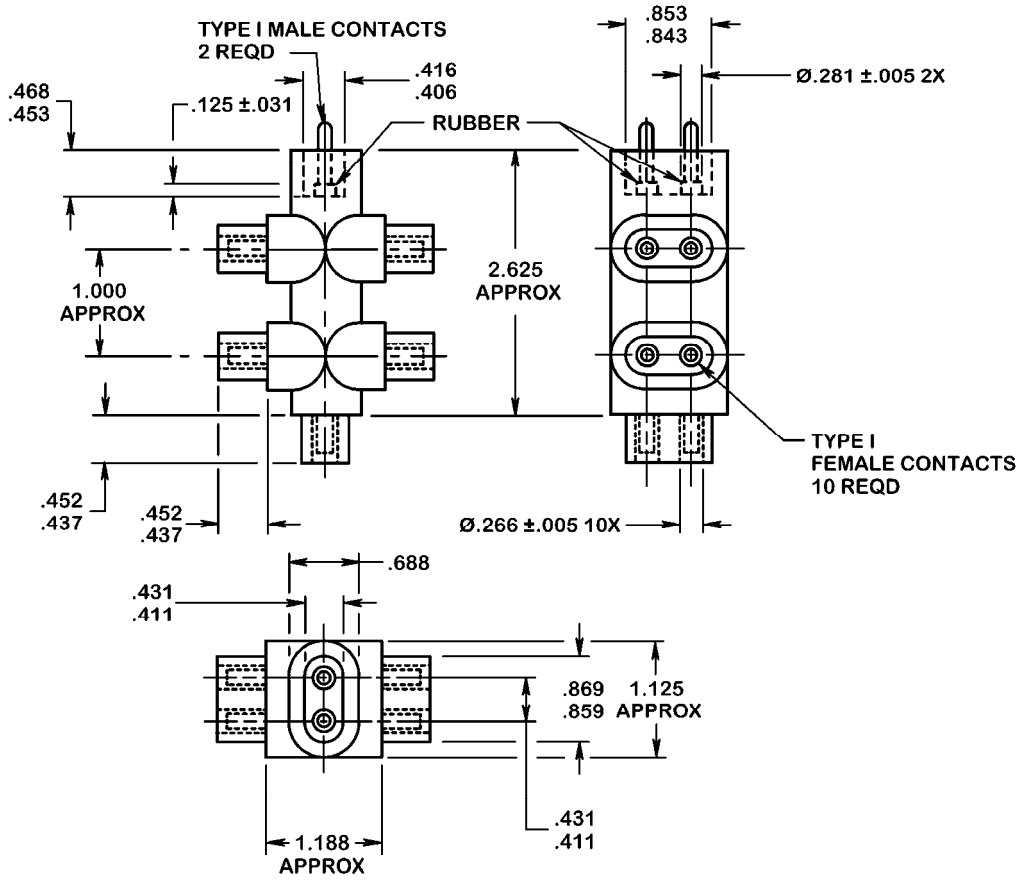
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NOTES:
1. Female contacts shall be recessed .125 ±.016 below face.
2. Connector rating: 600 volts, 15 amperes.
3. Voltage drop: Not more than 10 millivolts at 15 amperes.
4. Connecting and disconnecting force: 10-35 pounds.
5. All dimensions are in inches.

FIGURE 45. Adapter, cable assembly, three-way.

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NOTES:
1. Female contacts shall be recessed .125 ±.016 below face.
2. Connector rating: 600 volts, 15 amperes.
3. Voltage drop: Not more than 10 millivolts at 15 amperes.
4. Connecting and disconnecting force: 10-35 pounds.
5. All dimensions are in inches.

FIGURE 46. Adapter, cable assembly, five-way.

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Custodians:
Army - CR4
DLA - GS

Preparing Activity:
DLA - GS8

(Project 6150-2010-002)

Review Activity:
Army - AT

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